

White matter micro- and macrostructure brain charts for the human lifespan

<https://doi.org/10.1038/s41586-026-10454-2>

Received: 9 May 2025

Accepted: 26 March 2026

Published online: 13 May 2026

Open access

 Check for updates

Michael E. Kim¹✉, Chenyu Gao², Karthik Ramadass^{1,2}, Nancy R. Newlin¹, Praitayini Kanakaraj¹, Sam Bogdanov³, Gaurav Rudravaram², Derek Archer^{4,5,6}, Timothy J. Hohman^{4,5}, Angela L. Jefferson^{4,7,8,9,10}, Victoria L. Morgan^{11,12,13}, Alexandra Roche¹², Dario J. Englot^{1,2,11,13,14}, Susan M. Resnick¹⁵, Lori L. Beason-Held¹⁵, Laurie E. Cutting^{6,16}, Laura A. Barquero¹⁶, Micah A. D'archangel¹⁶, Tin Q. Nguyen^{6,12,16}, Kathryn L. Humphreys⁹, Yanbin Niu⁹, Sophia Vinci-Booher⁹, Carissa J. Cascio¹⁷, The HABS-HD Study Team*, L. Taylor Davis¹¹, Zhiyuan Li¹⁸, Simon N. Vandekar¹⁹, Panpan Zhang^{4,19}, John C. Gore^{11,12,13}, Bennett A. Landman^{1,2,6,8,9,10,11,12,13,19}✉ & Kurt G. Schilling^{2,11,12}✉

The human brain relies on a complex network of connections to function, with white matter acting as the primary communication highway between different brain regions^{1,2}. Disruptions in these critical communication pathways are linked to several neurological, psychiatric and developmental disorders^{3,4}. Although clinicians have long used standard growth charts to track physical development⁵, with more recent work translating these to whole-brain and grey matter measurements^{6–9}, there has been no equivalent reference standard for white matter. Establishing a readily available normative reference is an imperative first step if we hope to utilize these white matter structural biomarkers clinically. Here we present lifespan reference charts for human brain white matter. By processing and standardizing 35,120 brain scans from diverse global studies, we mapped the typical growth, maturation and age-related decline of specific brain pathways from birth to 100 years of age. These reference charts establish a fundamental benchmark for healthy brain development and ageing, allowing researchers and clinicians to quantify how an individual's brain deviates from typical patterns and highlighting disorder-related alterations. Furthermore, the accompanying open access charts enable the scientific and clinical communities to evaluate new patient and research data against these normative baselines, facilitating future clinical and neuroscience studies.

The human brain is an intricately organized network, with white matter (WM) forming the backbone of large-scale neural communication^{1,2}. Although much focus in neuroscience has been placed on grey matter morphology and its association with cognition, behaviour and disease, WM comprises nearly half of total brain volume and serves as the fundamental conduit for information transfer between cortical and subcortical regions. Disruptions in WM integrity are implicated across a spectrum of neurodevelopmental, neuropsychiatric and neurodegenerative disorders^{3,4}. However, despite these crucial roles of WM in brain functions, no standardized normative reference data have previously been derived for characterizing individual variability, developmental milestones or pathological deviations in WM structure. These may now be derived using appropriate image

analysis tools by leveraging the large array of imaging datasets that are publicly available.

The concept of normative brain charts—analogue to paediatric growth charts—has recently been introduced⁶ to benchmark some neuroanatomical trajectories across the lifespan, including grey matter features such as cortical thickness and volume^{6–9}. These efforts have revealed fundamental principles of neurodevelopment and ageing, enabling robust comparisons across individuals and clinical populations. However, in the context of WM, these previous efforts were predominantly limited to global volumetric analyses and did not provide the tract-specific, pathway-level phenotypes that are needed to localize deviations and to relate WM architecture to circuit function and dysfunction. This constraint is largely due to methodological challenges

¹Department of Computer Science, Vanderbilt University, Nashville, TN, USA. ²Department of Electrical and Computer Engineering, Vanderbilt University, Nashville, TN, USA. ³Medical Scientist Training Program, Vanderbilt University, Nashville, TN, USA. ⁴Vanderbilt Memory and Alzheimer's Center, Vanderbilt University Medical Center, Nashville, TN, USA. ⁵Vanderbilt Genetics Institute, Vanderbilt University Medical Center, Nashville, TN, USA. ⁶Vanderbilt Brain Institute, Vanderbilt University, Nashville, TN, USA. ⁷Department of Medicine, Vanderbilt University Medical Center, Nashville, TN, USA. ⁸Department of Neurology, Vanderbilt University Medical Center, Nashville, TN, USA. ⁹Department of Psychology and Human Development, Vanderbilt University, Nashville, TN, USA. ¹⁰Department of Psychiatry and Behavioral Sciences, Vanderbilt University Medical Center, Nashville, TN, USA. ¹¹Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center, Nashville, TN, USA. ¹²Vanderbilt University Institute of Imaging Science, Nashville, TN, USA. ¹³Department of Biomedical Engineering, Vanderbilt University, Nashville, TN, USA. ¹⁴Department of Neurological Surgery, Vanderbilt University Medical Center, Nashville, TN, USA. ¹⁵Laboratory of Behavioral Neuroscience, National Institute on Aging, National Institutes of Health, Baltimore, MD, USA. ¹⁶Department of Special Education, Peabody College of Education and Human Development, Nashville, TN, USA. ¹⁷Life Span Institute and Department of Psychology, University of Kansas, Lawrence, KS, USA. ¹⁸Department of Computer Science, Park University, Parkville, MO, USA. ¹⁹Department of Biostatistics, Vanderbilt University Medical Center, Nashville, TN, USA. *Lists of authors and their affiliations appear at the end of the paper. ✉e-mail: michael.kim@vanderbilt.edu; bennett.landman@vanderbilt.edu; kurt.g.schilling.1@vanderbilt.edu

of diffusion MRI (dMRI), the only non-invasive imaging modality that allows for studying directionality and organization of WM pathways in the human brain. These challenges include complexities in fibre tractography, inter-site variability and historically limited large-scale, harmonized datasets¹⁰. Given the increasing recognition of the role of WM in normal brain functions and in disease, establishing a rigorous, data-driven, normative reference for WM is a pressing scientific need. Just as with paediatric growth charts, defining this standard of healthy maturation is a necessary prerequisite for identifying and quantifying atypical development and decline^{11,12}.

dMRI provides a rich array of information on WM structure, including microstructural indices that reflect axonal density and dispersion¹³, as well as macrostructural properties such as WM tract volume, length and shape¹⁴. By integrating methodological advances in tract segmentation, data harmonization and statistical modelling that charts the full population distribution (for example, location, scale and skew) of inter-individual variability, we have established age- and sex-stratified trajectories of WM organization from early childhood to advanced ageing. By establishing standardized benchmarks for WM maturation and degeneration, we provide the foundational data to support discovery in fundamental neuroscience and for clinical translation, facilitating individualized assessments of brain health. These reference charts, along with openly accessible processing tools and harmonization pipelines, create a robust foundation for future research, large-scale collaborations and personalized approaches to studying WM alterations in health and disease.

We report our development of WM brain charts, a large-scale normative framework that documents changes in WM microstructure and macrostructure across the human lifespan. By integrating and analysing dMRI data spanning 50 different population studies, we have systematically characterized axonal density and dispersion, as well as tract volume, length and shape, in functionally relevant WM pathways, from early life to mature adulthood. These WM brain charts may be used in several important applications including: (1) defining normative trajectories of microstructural and macrostructural features of specific WM pathways; (2) revealing previously uncharacterized developmental milestones, refining our understanding of critical periods of WM growth and decline; (3) enabling sensitive detection of deviations from normative pathways, with implications for identifying early markers of neurodevelopmental and neurodegenerative disorders; and (4) enabling standardized centile scoring of new datasets to quantify WM alterations in clinical populations.

Mapping normative WM growth

We included and analysed dMRI data from a cohort of 35,120 individuals spanning 50 population studies and including 4,253,251 imaging volumes, representing typical development and ageing with no known neurological or psychiatric conditions. We modelled both global and tract-specific WM phenotypes—including 72 anatomically defined pathways—using generalized additive models for location, scale and shape (GAMLSS)¹⁵, a flexible statistical framework endorsed by the World Health Organization for modelling non-linear biological trajectories¹⁶. GAMLSS enabled simultaneous estimation of age-dependent changes in location, scaling and skewness, while accounting for study-level batch effects. Microstructural (fractional anisotropy (FA), mean diffusivity (MD), axial diffusivity (AD) and radial diffusivity (RD)) and macrostructural (tract volume, length and surface area) features were modelled across the full lifespan (Fig. 1 and Extended Data Table 1; demographic information is presented in Supplementary Table S2.T1 and Supplementary Table S2.T9).

Lifespan trajectories of global WM features revealed distinct, feature-specific patterns of growth and decline (Fig. 1b,c). Cerebral WM volume increased rapidly during early development, peaking at 34 years, followed by gradual atrophy in later life. FA exhibited a similar,

though slightly earlier, inflection—increasing during childhood and adolescence, peaking around age 24, then steadily declining. By contrast, diffusivity measures (MD, RD and AD) demonstrated inverted trajectories: decreasing steeply during early life, reaching their lowest points in middle adulthood (RD at approximately 32 years, MD at approximately 40 years and AD at approximately 46 years), followed by progressive increases into older age. These divergent trajectories delineate critical inflection points in WM maturation and degeneration.

GAMLSS-estimated population variability revealed distinct age-related trends across WM phenotypes (Fig. 1d). Variability in metrics of diffusivity remained low or declined during early life, followed by substantial increases during mid-to-late adulthood. FA variability peaked during early childhood, declined steadily through midlife, and increased again in older age—suggesting age-dependent fluctuations in structural heterogeneity. Rates of change (Fig. 1e) were steepest during early childhood, reflecting rapid developmental transitions, and gradually diminished through late childhood and early adolescence (ages 6–12), mirroring the extended maturation period of myelination and axonal remodelling described in histological and neuroimaging studies^{17–19}.

Tract-specific WM phenotypes

To move beyond global cerebral measures and characterize regional heterogeneity, we applied the GAMLSS framework to derive tract-specific normative trajectories for WM microstructural and macrostructural features across the lifespan. We estimated trajectories for 72 anatomically defined WM pathways, capturing the full spectrum of association, commissural, projection, thalamic and limbic systems (see Extended Data Table 2 for a list of tracts and abbreviations). Representative trajectories are shown for five exemplar tracts: the arcuate fasciculus and cingulum bundle (association), corticospinal tract and anterior thalamic radiation (projection), and genu of the corpus callosum (commissural) for microstructural phenotypes (Fig. 2 and Extended Data Figs. 1–3; Supplementary Fig. S1.F1 for contralateral) and macrostructural phenotypes (Fig. 3 and Extended Data Figs. 4 and 5; Supplementary Fig. S1.F2 for contralateral; Supplementary Figs. S1.F3–S1.F11 for trajectories of normalized measures). Selection of visualized tracts is based on prevalence of study in the literature²⁰ (and when displayed in figures are coloured to highlight directionality, with red, green and blue indicating left–right, anterior–posterior and superior–inferior orientations, respectively)²¹.

Tract-wise trajectories of FA revealed consistent developmental phases but variable temporal profiles across pathways (Fig. 2). FA increased rapidly during infancy and childhood, reached a plateau in early adulthood, and declined gradually in later life. Yet the magnitude and timing of these inflection points varied substantially by tract. For instance, the corticospinal tract exhibited a prolonged plateau of high anisotropy into midlife, and the genu of the corpus callosum and cingulum bundle reached peak FA earlier, followed by earlier declines. Inter-individual variability in FA followed a similar pattern, remaining low and stable during early development, then increasing with age in a tract-dependent manner. First-derivative estimates highlighted that FA change rates were steepest in early development but differed in timing and slope across pathways.

Macrostructural trajectories of tract volume followed characteristic inverted U-shaped curves (Fig. 3), with rapid volumetric expansion during early development, peaks occurring in adolescence or early adulthood, and progressive atrophy in older age. As with microstructure, these patterns were highly tract-dependent, with marked differences in peak timing (Supplementary Fig. S4.F8) and slope of decline (Supplementary Table S2.T9). Inter-individual variability in volume increased with age, and the steepest rates of volumetric change occurred during early childhood. However, the timing and magnitude of these dynamics varied across tracts, reflecting anatomical and developmental diversity in WM maturation and degeneration. Furthermore, although sex differences in microstructural measurements are less pronounced

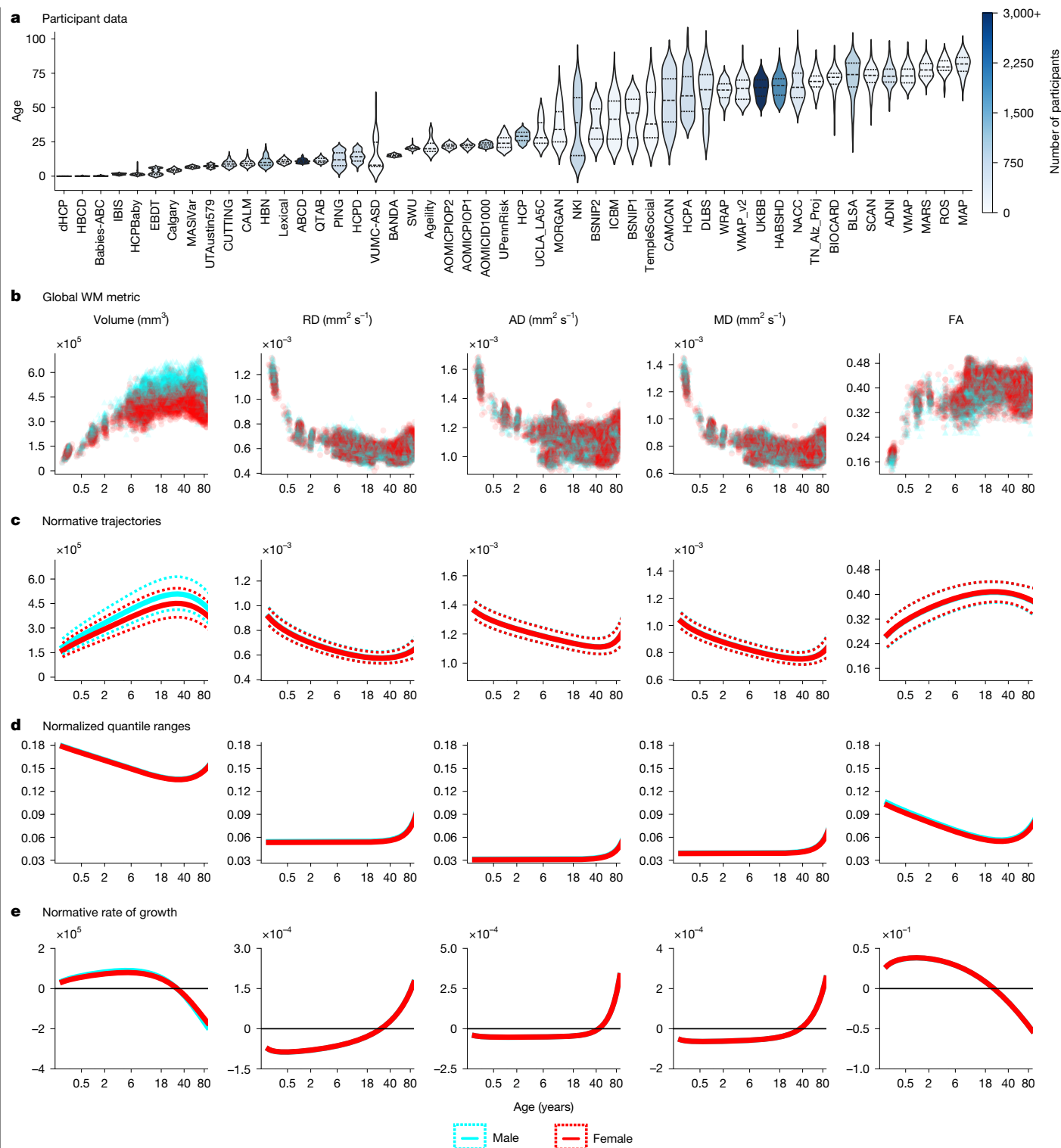


Fig. 1 | Global WM brain charts across the human lifespan. **a**, Data were aggregated from 50 imaging studies, comprising $n = 35,120$ scans from typically developing participants aged 0 to 100 years, forming the basis for the lifespan WM brain charts. Violin plots show the age distribution for each study coloured by the number of typically developing or ageing participants. **b**, Raw data points illustrate the distribution of global WM metrics (volume, FA, MD, AD and RD) across the lifespan for male (cyan) and female (red) participants. **c**, Normative trajectories for each metric, modelled using GAMLSS, show the median (solid lines) and 2.5th and 97.5th percentiles (dotted lines). **d**, Normalized quantile

ranges (difference between 25th and 75th percentiles divided by the median) depict changes in population variability across the lifespan for each global WM feature. **e**, The normative rate of change ($d/d(\log(\text{age}))$) for the median centile visualizes the velocity of WM development with age, highlighting that growth and decline are most rapid early in life and that different WM features reach their peak or minimum values at distinct ages. The horizontal line ($y = 0$) denotes critical points at which the derivative of the trajectories change sign. Note that x axes are log-scaled to emphasize developmental and ageing periods.

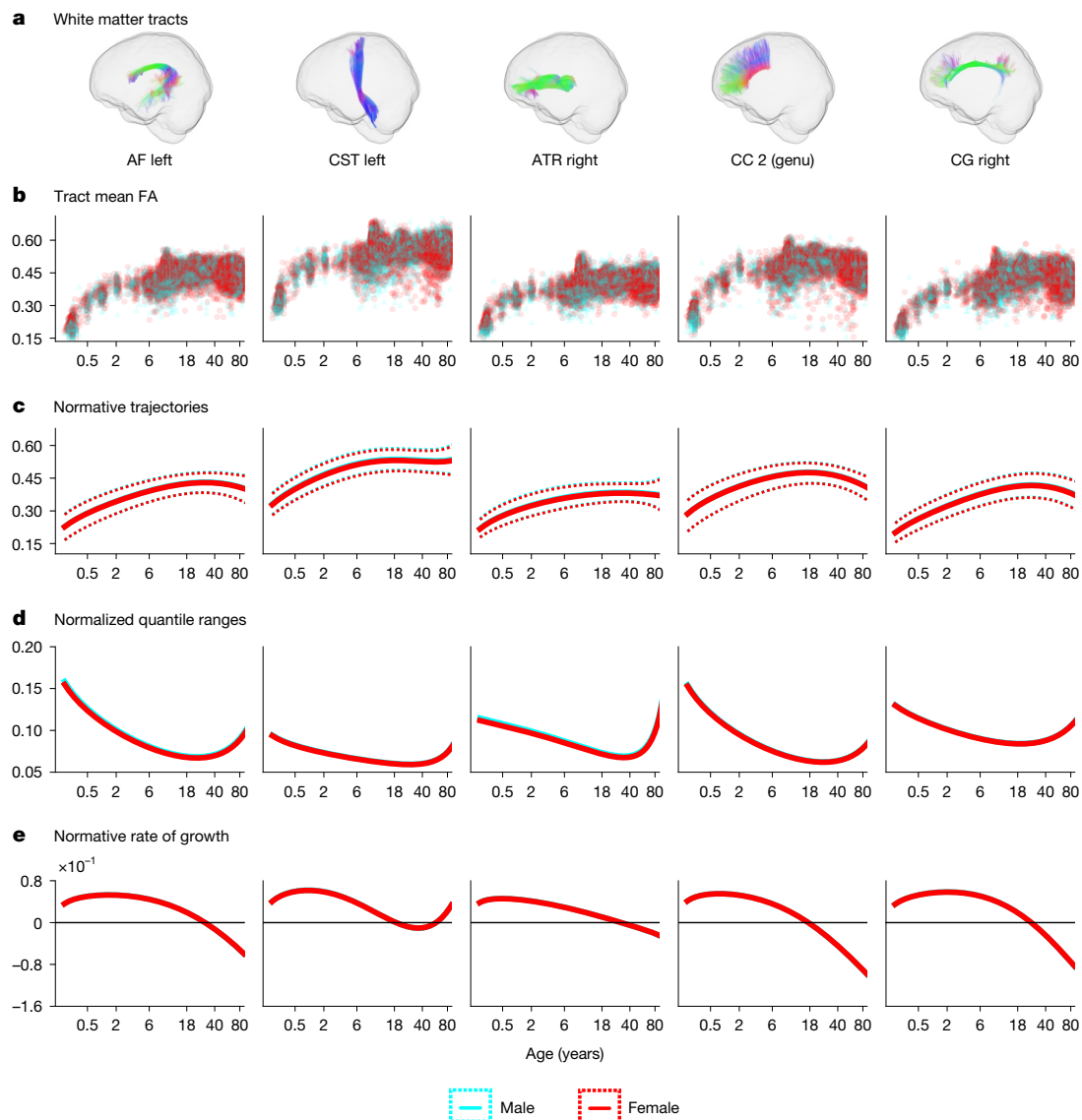


Fig. 2 | Tract-specific microstructural FA brain charts. Lifespan FA brain charts for WM microstructure (see Extended Data Figs. 1–3 for MD, AD and RD; see Supplementary Fig. S1.F1 for contralateral) reveal distinct developmental trajectories and variability across different WM pathways. **a**, Five exemplar tracts are shown: left arcuate fasciculus (AF left), left CST (CST left), right anterior thalamic radiation (ATR right), genu of the corpus callosum (CC 2 (genu)) and the right cingulum (CG right). Tracts are coloured by orientation, with blue indicating inferior–superior, red indicating left–right, and green indicating posterior–anterior. Graphs in **b–e** are aligned with the corresponding tracts. **b**, Raw FA data points for these tracts extend across the lifespan.

c, Normative GAMLSS trajectories show FA increasing during development, plateauing in adulthood and declining in later life, where the timing and magnitude vary by tract. Median (solid lines) and 2.5th and 97.5th percentiles (dotted lines) are shown. **d**, Normalized quantile ranges indicate that FA variability is generally lowest in middle age and increases later in life, although developmental variability patterns differ between tracts. **e**, The normative rate of change ($d/d(\log(\text{age}))$) for the median centile suggests that FA peaks at different ages and changes at different rates depending on the specific tract. Note that x axes are log-scaled to emphasize developmental and ageing periods.

than in macrostructure, we still observe significant effects for both (Supplementary Fig. S4.F6).

Together, these tract-specific brain charts delineate the heterogeneity of WM development and ageing, capturing normative ranges, variability and rates of change across anatomically distinct pathways. This reference framework enables precise interpretation of age-related trajectories at the pathway level and establishes a foundation for identifying atypical patterns in both developmental and clinical contexts.

WM developmental milestones

We extracted tract-resolved milestones from lifespan centile trajectories by locating critical points (local extrema) for microstructural and macrostructural features (Fig. 4a). Examples from the left

arcuate fasciculus and right cingulum illustrate two points that generalize: (1) feature-specific timing differs within a tract; and (2) timing differs across tracts. Across pathways, macrostructural metrics (length, surface area and volume) typically peak earlier than microstructural metrics (Supplementary Fig. S4.F8). The bottom context panel orients the trajectories within external timelines—marking cohort age coverage, literature-based diagnostic ages for the major disorders represented²² and developmental transcriptomic windows²³—highlighting periods of pronounced gene-expression change and clinical vulnerability.

We then used the charts to investigate *in vivo* microstructural changes that reflect myelination. To relate milestones to myelination, we used RD²⁴ as a myelin-sensitive proxy and segmented each RD trajectory piecewise (Methods, ‘WM lifespan milestones’). This captures the early life decrease (consistent with progressive myelination) and late-life

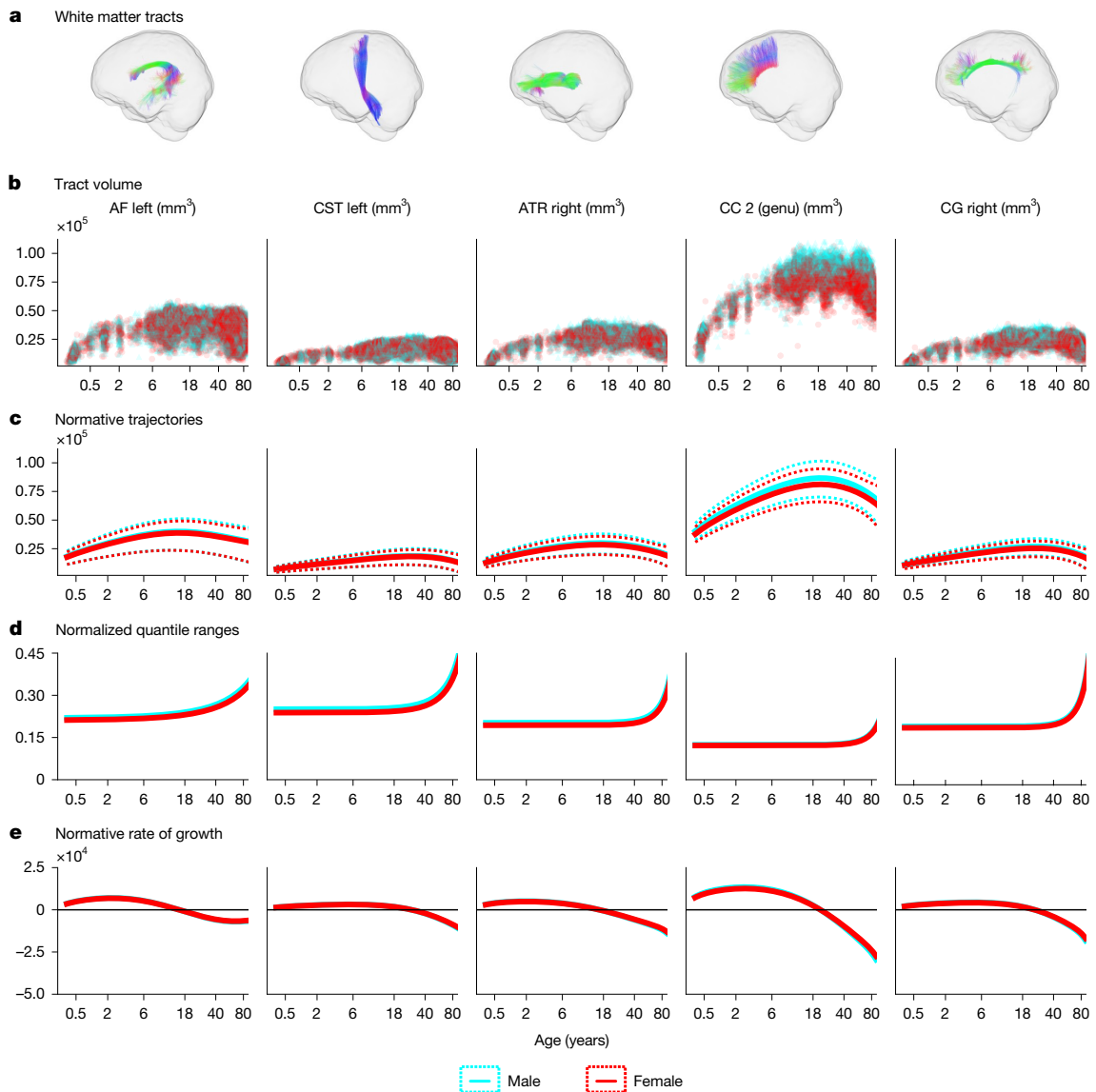


Fig. 3 | Tract-specific macrostructural volume brain charts. Lifespan brain charts of pathway volume for WM macrostructure (see Extended Data Figs. 4 and 5 for length and surface area and Supplementary Fig. S1.F2 for contralateral) reveal distinct developmental trajectories and variability across different WM pathways. **a**, Five exemplar tracts are shown: AF left, CST left, ATR right, CC 2 (genu) and CG right, as in Fig. 2a. Tracts are coloured by orientation, with blue indicating inferior–superior, red indicating left–right, and green indicating posterior–anterior. Graphs in **b–e** are aligned with the corresponding tracts. **b**, Raw tract volume data points for these tracts span the lifespan, illustrating differences in typical volume ranges between tracts. **c**, Normative GAMLSS

trajectories show tract volume increasing during development, peaking in adolescence or early adulthood and declining in later life, with tract-specific timing and magnitude. Median (solid lines) and 2.5th/97.5th percentiles (dotted lines) are shown. **d**, Normalized quantile ranges indicate that volume variability increases later in life and differs between sexes, and is generally lowest in middle age. **e**, The normative rate of change ($d/d(\log(\text{age}))$) for the median centile suggests that tract volume peaks at different ages and the rate of subsequent decline varies depending on the specific tract. Note that x axes are log-scaled to emphasize developmental and ageing periods.

increases in RD, with tract ordering that mirrors classical staging^{18,25,26}: earlier for projection systems (for example, corticospinal tract (CST) and frontopontine tract (FPT)) and later for association pathways (for example, uncinate fasciculus (UF))^{27,28} (Fig. 4b,c). These patterns are in line with the established view that human myelination is protracted into mid-adulthood and proceeds in non-simultaneous waves across systems rather than as a single schedule^{18,29}. We summarize the heterogeneity in timing and dynamics across pathways (Fig. 4c), reporting ages of key RD time points intended to reflect distinct waves of myelination²⁵ (end of early rapid myelination, maturation throughout adolescence and onset of myelin decline) alongside the rates of change between each time point. We emphasize RD as a sensitive, but non-specific in vivo proxy that can reflect myelination as well as other microstructural factors¹³.

Because the charts yield tract-specific milestones, they also enable quantitative assessment of links between maturation and later life change (Fig. 4d). Under a retrogenesis framework³⁰, we evaluated two tract-level hypotheses derived from centile trajectories. First (based on relative timings), the ‘last-in, first-out’ hypothesis predicts that later maturing pathways show earlier onset of decline³¹; operationally, we related the age of each tract at feature maturity to the age at which decline begins (illustrated for RD). We did not find evidence supporting the last-in, first-out hypothesis in these timing estimates; instead, later maturing pathways tended to show later onset of decline. Second (based on relative rates of change), the ‘gain-predicts-loss’ hypothesis suggests that pathways with larger developmental gains should show steeper ageing-related losses¹⁷; operationally, we related adolescent

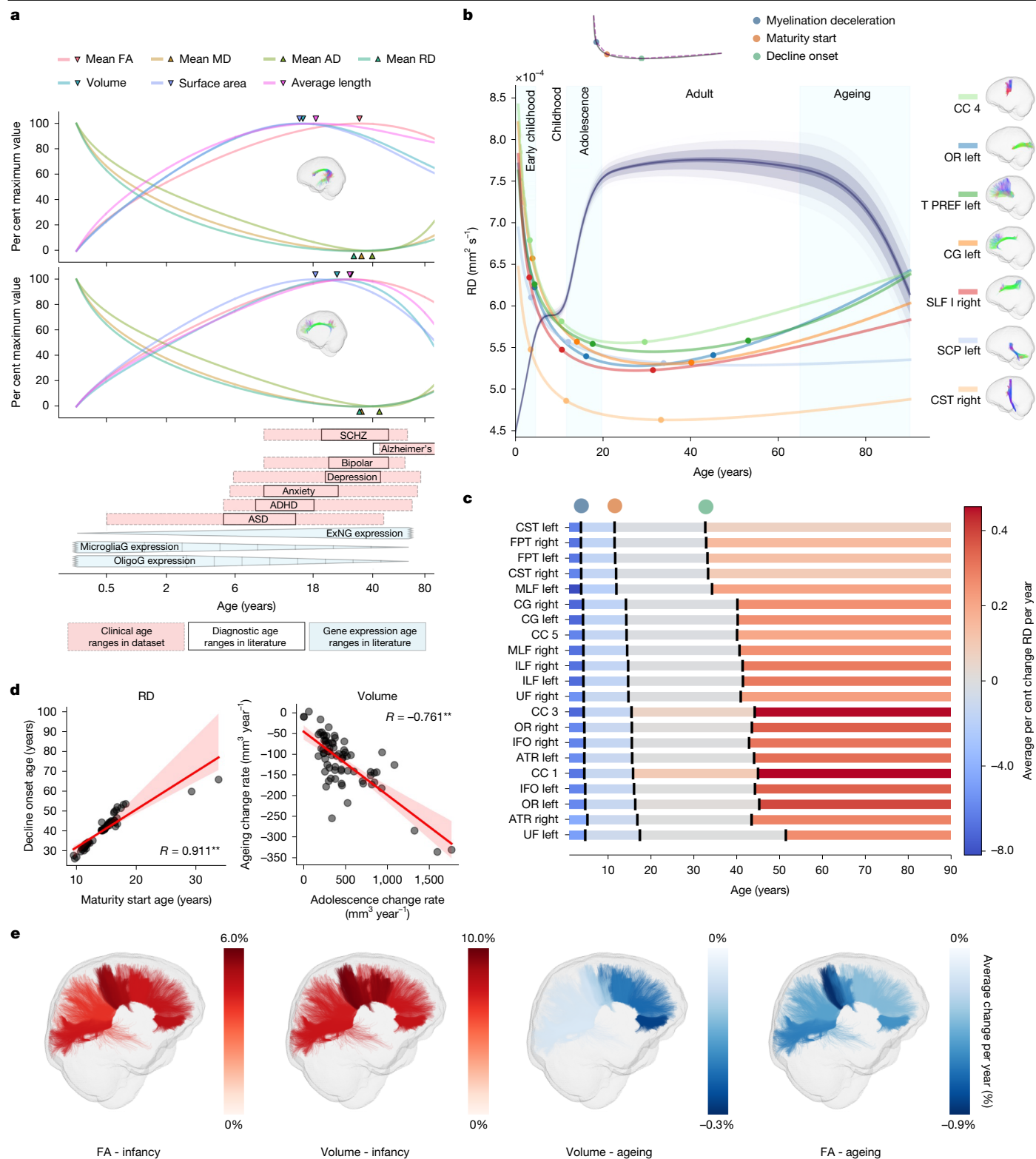


Fig. 4 | See next page for caption.

growth rates to later life decline rates (illustrated for tract volume). Macrostructural trajectories were consistent with this hypothesis, with faster adolescent volumetric expansion being associated with steeper volumetric decline in ageing (parallel regressions for additional WM features shown in Supplementary Fig. S4.F9). Together, these results indicate that centile trajectories capture systematic, feature-dependent couplings between maturation and later life change.

Finally, the charts localize spatial organization of WM change³² (Fig. 4e). We summarize the mean rate of change for FA and volume in infancy and ageing along the sections of the corpus callosum. During infancy, microstructural rates show an anterior–posterior ‘inside-out’ trend, with relatively slower change at the most anterior and posterior pathways and faster change in intermediate systems. During ageing, front-to-back volumetric differences emerge with stronger anterior

Fig. 4 | Tract-specific developmental milestones, neurobiological patterns and spatial gradients. Trajectories reveal developmental milestones, enabling investigation of neurobiological hypotheses and spatio-temporal microstructural dynamics. **a**, Median centile trajectories for all features of the left arcuate fasciculus (top) and the right cingulum (middle) show pathway-specific developmental maturation as peaks (downward triangles) or troughs (upward triangles). Bottom, a graphical summary situates these trajectories alongside non-MRI milestones. Pink boxes denote the age range for clinical disorders in the dataset; black boxes denote literature diagnostic ranges²²; blue tapered bars indicate transcriptomic development windows²³. ADHD, attention-deficit hyperactivity disorder; ASD, autism spectrum disorder; SCHZ, schizophrenia; ExNG, excitatory neuron gene; MicrogliaG, microglia gene; OligoG, oligodendrocyte gene. **b**, Median RD trajectories for exemplar tracts illustrate waves of development and late-life change (waves identified by piecewise segmentation of lifespan curves (top inset)). These in vivo patterns are consistent with proposed myelin development and ageing patterns²⁵ based on foundational histological work from Yakovlev and Lecours¹⁸. Myelin changes are visualized qualitatively as the purple curve (adapted from ref. 25, Springer Nature). OR, optic radiation;

SCP, superior cerebellar peduncle; SLF, superior longitudinal fascicle; T PREF, thalamic prefrontal. **c**, Relative timing and magnitude of change for RD trajectories across selected tracts. Vertical black lines represent the identified myelination time points (for example, rapid early myelin increase, maturity start and decline onset). Bar colour represents average annual per cent change during life epochs, revealing tract-specific variability in rates of change. IFO, inferior occipito-frontal fascicle; ILF, inferior longitudinal fascicle; MLF, middle longitudinal fascicle. **d**, Neurobiological hypothesis tests derived from tract milestone timings (left) and rates (right), with each datapoint representing one tract. Left, there is a positive relationship between age at feature maturity and age at decline onset for RD. Right, greater adolescent growth in macrostructure tract volume predicts steeper decline with ageing (that is, gain predicts loss; slope–slope relation). ** indicates significance of linear regression under Bonferroni correction ($P < 0.0035$; the shaded area represents 95% confidence intervals; additional feature regressions are shown in Supplementary Fig. S4.F9). **e**, Spatial organization. Anterior-to-posterior gradients emerge in rates of change across tracts (both microstructure and macrostructure), indicating regionally patterned maturation and decline.

declines. These gradients differ by feature and life stage and are not a mirror image of development, indicating that retrogenesis provides a partial, but incomplete account of the spatial organization of change.

These results collectively define a fine-grained atlas of WM developmental milestones. The timing, variability and rates of change in both microstructural and macrostructural features differ substantially across pathways, offering a comprehensive reference for interpreting normative WM maturation and age-related decline. This framework supports future efforts to investigate how developmental timing relates to functional specialization, vulnerability to disease and lifespan trajectories of brain structure.

Individualized tract centile scores

We computed individualized centile scores to benchmark WM measurements of each participant against normative age- and sex-stratified trajectories (Methods, ‘Centile scores and case–control differences’). These scores quantify the relative typicality or atypicality of microstructural and macrostructural features on a unified scale, with values near the extremes (0th or 100th percentile) reflecting substantial deviation from the normative population. Leveraging a clinically diverse dataset, we applied this framework to systematically examine case–control differences across several diagnostic groups, including neurodevelopmental, neurocognitive and neurodegenerative disorders (Fig. 5 and Extended Data Table 3).

To summarize deviations across multiple features and tracts, we computed a normalized centile Mahalanobis distance (nCMD) for each individual, capturing aggregate atypicality within microstructural, macrostructural and combined feature spaces (Methods, ‘Centile scores across cognitive groups’). Across diagnostic groups, nCMD values were consistently higher relative to controls (Fig. 5a; see Supplementary Table S3.T1 for effect sizes). This composite score enabled dimensionality reduction while preserving individual variation, revealing disorder-specific patterns—some of which were characterized primarily by microstructural deviation, whereas others were characterized by macrostructural or mixed profiles.

Significant deviations from normative centile scores were also observed for specific tract microstructure and macrostructure measurements. Diagnostic groups of mild cognitive impairment (MCI) and Alzheimer’s disease showed significant deviations across most tracts and features compared with neurotypical controls (Fig. 5b,c; see Extended Data Figs. 6 and 7 and Supplementary Fig. S3.F1 for other diagnostic groups, Supplementary Figs. S3.F3–S3.F5 for alternative false discovery rate with Benjamini–Hochberg correction and Supplementary Figs. S3.F6–S3.F8 for effect sizes). MCI was associated with

widespread increases in diffusivity, reductions in FA and lower tract volumes and surface areas—changes that were even more pronounced in Alzheimer’s disease.

These findings demonstrate that individualized centile scores offer a sensitive, interpretable framework for quantifying WM atypicality—capturing both global deviations and pathway-specific abnormalities across diverse neurological diagnostic groups.

Centile scoring of new datasets

A central challenge in applying normative brain charts is estimating individualized centile scores for new, out-of-sample (OOS) datasets that are not included in the original reference cohort. To address this, we implemented a maximum likelihood estimation (MLE) framework to align external datasets to existing normative trajectories on a feature and pathway-specific basis (Fig. 6 and Methods, ‘OOS centile estimation’). This alignment enables accurate and interpretable centile scoring, thereby extending the utility of WM brain charts to new datasets for studying both typical and pathological brain structure.

As an initial validation, we estimated study-specific offsets using cognitively healthy controls from a held-out dataset, yielding a near-uniform centile distribution centred around the population median (50th centile; Fig. 6a). This enabled two key applications in a new Alzheimer’s disease cohort. First, aligned centile scores provided a consistent framework for evaluating case–control differences, revealing specific WM features and pathways that significantly diverged in Alzheimer’s disease relative to controls (Fig. 6b). Second, the centile framework enabled associations between WM structure and cognitive performance to be examined, identifying pathway-specific features that were strongly linked to clinical variability (Fig. 6c). In this way, the reference charts can be used to relate abnormality indices to a broad range of external clinical data.

Together, these findings demonstrate that centile scoring of OOS datasets provides a rigorous and generalizable framework for evaluating group differences, identifying pathway-specific deviations and exploring structure–function relationships. By grounding these analyses in normative WM trajectories, our approach enables biologically interpretable comparisons across diagnostic groups and datasets (see Supplementary Information, section 8 for how to perform alignment with an OOS dataset).

Discussion

Here we present normative brain charts of WM microstructure and macrostructure across the human lifespan, derived from 35,120 dMRI

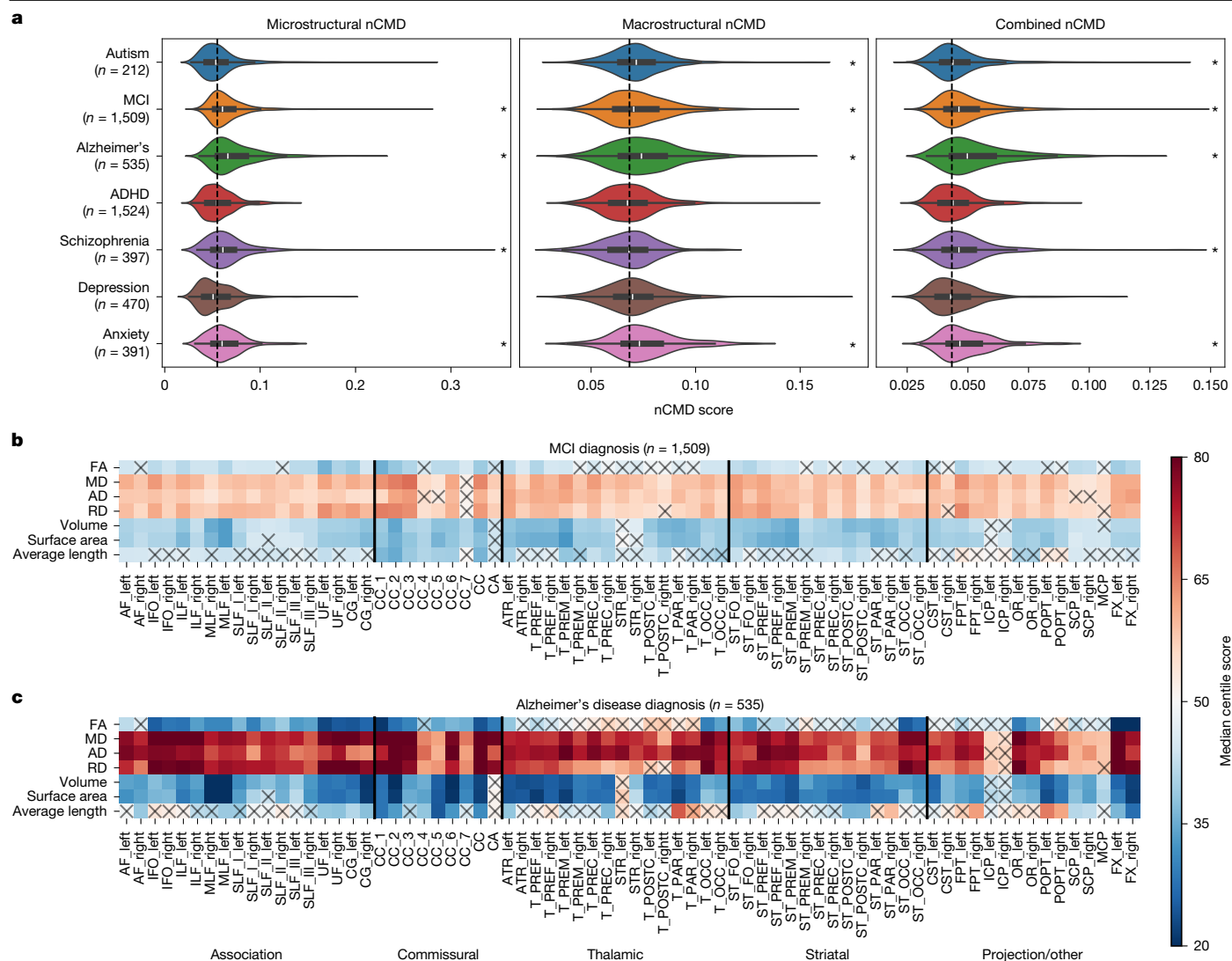


Fig. 5 | Case-control differences quantified by WM centile scores. Deviations from normative WM trajectories, measured using individualized centile scores (ranging 0–100, with 50 being the population median), reveal significant differences between typically developing controls and patient populations with diverse neurological conditions. **a**, Violin plots show distributions of the nCMD scores, an aggregate measure of deviation across all tracts, for microstructural, macrostructural and combined features in several diagnostic groups. Internal box plots show the median (central white line), with the bounds of the box representing 25th and 75th percentiles (interquartile range (IQR)) and whiskers extending to the maxima and minima within $1.5 \times$ IQR. Higher nCMD scores indicate greater overall deviation from the normative control population (compared against median nCMD for typically developing and ageing participants, shown as vertical dashed lines). Significant differences from controls (Bonferroni-corrected two-sided one-sample Wilcoxon signed-rank test) are marked with an asterisk, notably for MCI, Alzheimer's disease and

anxiety groups across all aggregated measures (significant at $P < 7.4 \times 10^{-3}$). **b, c**, Heat maps display the median centile score for each WM feature across all 72 tracts for individuals diagnosed with MCI (**b**) and Alzheimer's disease (**c**). Colours represent the median centile relative to the median of the control (blue < 50 , red > 50). Widespread deviations are evident, particularly lower FA and volume as well as higher diffusivities in MCI and Alzheimer's disease. Crosses indicate features for which the median centile was not significantly different from 50 after Bonferroni correction under a two-sided one-sample Wilcoxon signed-rank test (significant at $P < 9.92 \times 10^{-5}$). CA, anterior commissure; FX, fornix; ICP, inferior cerebellar peduncle; MCP, middle cerebellar peduncle; POPT, parieto-occipito-temporal; STR, superior thalamic radiation; ST_FO, striato-fronto-orbital; ST_OCC, striato-occipital; ST_POSTC, striato-postcentral; ST_PREC, striato-precentral; ST_PREF, striato-prefrontal; ST_PREM, striato-premotor; ST_PAR, striato-parietal; T_OCC, thalamic-occipital; T_PAR, thalamic-parietal; T_POSTC, thalamic-postcentral; T_PREC, thalamic-precentral.

scans across 50 harmonized cohorts. These charts establish normative references for both global and tract-specific WM properties, enabling age and sex-stratified benchmarking of WM development, ageing and pathology. Our analyses delineate normative trajectories, identify tract-specific growth milestones and demonstrate individualized quantification of WM deviations with applications to diagnostic groups. By addressing prior methodological and dataset limitations, this work marks a critical step toward standardized, lifespan quantification of WM organization.

First, our WM brain charts define robust, sex-stratified trajectories for both microstructural and macrostructural properties across 72 WM

pathways. This includes canonical diffusion metrics (FA, MD, AD and RD) and tract-specific morphometric features such as volume, length and surface area. Prior lifespan studies have focused largely on grey matter, and established normative models are now available for cortical and subcortical volumes (including BrainChart⁶, NiChart⁹ or Centile-Brain^{7,8} platforms). Whereas recent efforts have created normative models for WM microstructure based on regional diffusion metrics^{33,34}, our work extends these frameworks by incorporating both microstructural and macrostructural measures across 72 anatomically defined pathways and providing a more granular and comprehensive view of WM development, ageing and pathology. Our results show that WM

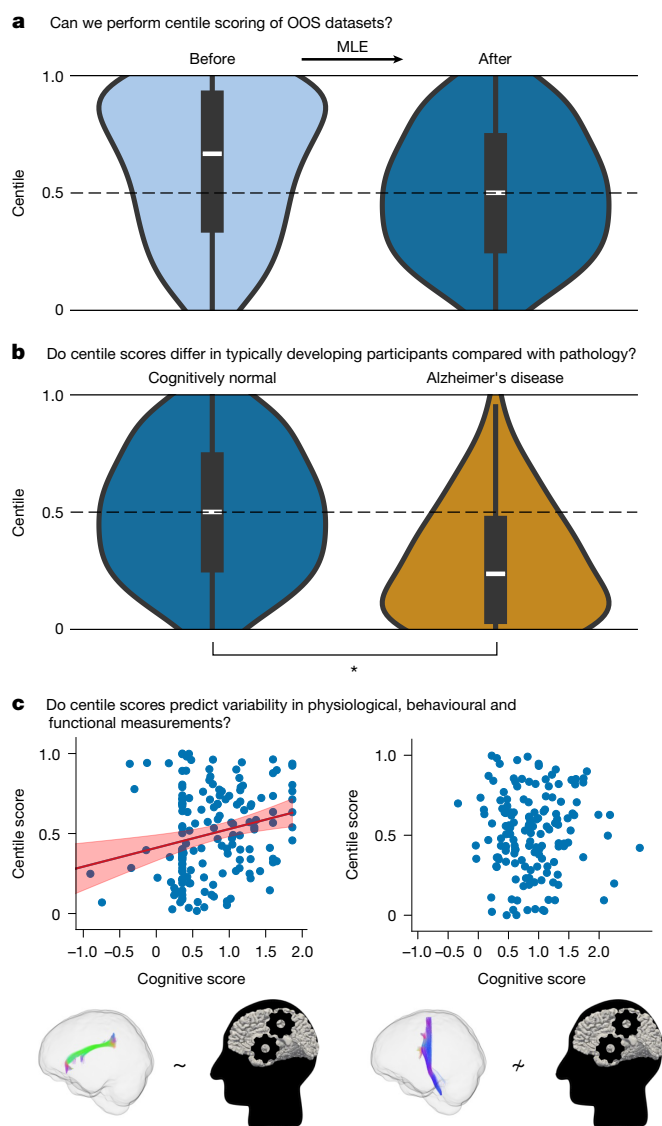


Fig. 6 | Applications of OOS centile scoring. Normative WM brain charts enable standardized centile scoring for new OOS datasets, facilitating harmonized analyses. Exemplar applications are demonstrated using the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset (withheld from the models used for this figure for demonstration purposes). **a**, OOS datasets can be aligned to the reference trajectories using MLE on typically developing or ageing participants to estimate study-specific offsets. This process yields a correctly centred centile distribution for OOS controls ($n = 591$ for the mean FA of CC1). Centile values before and after alignment are shown using a violin plot visualization with internal box plots showing the median (central white line), with the bounds of the box representing the IQR and whiskers extending to the maxima and minima within $1.5 \times$ IQR. The 50th centile is represented by a dotted black line. We recommend a sample size of $n \geq 100$ for alignment based on empirical assessments (see Supplementary Fig. S8.F1). **b**, Aligned centile scores allow for standardized comparisons between diagnostic groups within the OOS dataset, illustrated here comparing typically developing and ageing individuals (cognitively normal, $n = 591$) and those with Alzheimer's disease ($n = 212$). Asterisk denotes significance under a two-sided Mann-Whitney U test, with $P < 0.05$. **c**, Aligned centile scores serve as standardized metrics for investigating relationships with external variables; here we provide an illustrative cognition-specific example. Executive function (composite score⁴³) is significantly correlated with aligned FA centiles in the right SLF III (left; $n = 160$), but not the right CST (right; $n = 151$) for cognitively normal participants, highlighting pathway-specific associations. Significance was assessed for the centile-executive function association using a least-squares linear regression. Significance ($P < 6.94 \times 10^{-4}$ following Bonferroni correction) of the association is indicated with a red regression line and 95% confidence intervals.

pathways can be reliably characterized across development and ageing using harmonized tractography pipelines. Although diffusion tensor imaging (DTI)-derived microstructure has been well-studied^{28,33,35}, tract-level macrostructural features have received limited attention²⁷. Here we demonstrate their value and provide intuitive metrics, such as tract volume, average length and surface area, in both unnormalized and brain-size-normalized forms. Together, these benchmarks enable systematic characterization of WM maturation and support broader applications in developmental neuroscience and clinical research.

Second, our normative charts reveal pathway-specific developmental milestones, capturing the diversity in timing and sequence of maturation and decline across WM tracts. These differences reflect both functional specialization and distinct temporal dynamics of WM plasticity, preservation and degeneration²⁰. Furthermore, our normative charts agree with foundational¹⁸ and more recent works of myelin development across the lifespan, not only with the timings of myelination waves, but also with the relative myelination timings of certain WM tracts²⁵. The open-access trajectories generated here support testing of neurobiological theories^{17,31} such as the last-in, first-out hypothesis, where later-maturing pathways are more susceptible to early ageing, and the gain-predicts-loss hypothesis, which posits that regions undergoing rapid developmental expansion are prone to steeper age-related decline. The observed dissociation between microstructural and macrostructural inflection points further underscores the complex, feature-dependent nature of WM development, inviting future work into the temporal interplay of structural metrics across the lifespan.

Third, we demonstrate that normative WM brain charts derived from typically developing and ageing individuals with no known neurological or psychiatric conditions provide a sensitive and interpretable framework for detecting individualized anomalies in WM structure. Microstructural and macrostructural aberrations in cerebral WM have been implicated in numerous neurological disorders, as referenced in this work. Although we have been able to measure many of these parameters for over a decade, the lack of a reliable robust reference atlas has limited our ability to leverage these measurements as a tool for diagnosis and prognosis on an individual patient level. If we hope to utilize these measurements in a clinically productive manner, it is imperative to establish a readily available normative atlas of WM structural characteristics. Just as with any blood or body fluid laboratory test, a benchmark of normal values, ideally in a demographically similar population, is critical to interpreting the result in an individual patient. By providing a normative baseline, a single snapshot in time can be used to determine whether an individual exhibits an atypical phenotype relative to their age and sex. Crucially, the GAMLSS framework moves beyond solely a median trajectory and explicitly models the full population distribution (including its location, scale and shape), allowing quantification of inter-individual variability. By quantifying deviations from these distributions using centile scores, we identified distinct patterns of microstructural and macrostructural atypicality in individuals across a variety of diagnostic groups (Fig. 5, Extended Data Figs. 6 and 7 and Supplementary Figs. S3.F1–S3.F5). These results highlight the utility of centile-based assessments for mapping disease-related alterations at the level of specific tracts and features. Although mechanistic inferences are beyond the scope of this study, the sensitivity of this framework to detect subtle and widespread WM abnormalities underscores its potential for clinical applications in both diagnosis and tracking of neurological disorders.

Finally, our framework enables standardized effect size estimation and tract-level quantification of WM deviation in OOS datasets. By aligning new data to normative trajectories using maximum likelihood estimation, we enable centile scoring even in external cohorts collected across disparate sites and protocols. This facilitates cross-cohort comparisons, phenotype-feature associations and broader reproducibility across neuroimaging studies. As an illustrative example with Alzheimer's disease cohorts, these aligned centile scores provide a robust

foundation for exploring the relationship between WM features and clinical or cognitive outcomes. We note that as the brain charts were constructed with typically ageing and developing individuals, aligning OOS datasets requires some typically ageing and developing individuals in the OOS dataset as well (see Supplementary Information, ‘OOS alignment analysis’).

Several limitations warrant consideration. Although the aggregated dataset is among the largest to date, age distribution was uneven, with relative under-representation of infancy and mid-adulthood (Supplementary Fig. S2.F2). To maintain the statistical power of our normative reference model, we did not include phenotypes beyond sex (for example, ethnicity or genotype status) owing to limited data availability and reduced sample size (Supplementary Table S2.T1). Further, normative trajectories were derived from cross-sectional data; future longitudinal validation will be important, as individual trajectories can differ from population-level trends³⁶. Our cross-sectional approach was chosen to ensure statistical power and model stability for OOS alignment, as detailed in Supplementary Information, ‘Considerations for cross-sectional versus longitudinal brain charts’. Although there was data representation from several continents, the datasets used to create the brain charts are predominantly from European and North American populations (Supplementary Table S2.T2). This bias is unfortunately common in neuroimaging and genetics, echoing limitations that are found even in foundational clinical references such as the US Centers for Disease Control and Prevention and World Health Organization growth charts¹². This highlights a critical need for the global scientific community to increase ethnic, socioeconomic and demographic diversity in future MRI research. In our assessments relating developmental and ageing milestones for myelination and other properties, we note there are limitations in defining discrete developmental time points across continuous curves. Tractography-based metrics are inherently sensitive to preprocessing parameters and may be less reliable in certain populations or developmental stages^{10,37}. Future work may also include alternative bundle segmentation methods¹⁴, with varying levels of specificity and sensitivity³⁸, alternative definitions and interpretations of pathways, or under-investigated pathways of the brainstem or short association fibres²⁶. Although we utilized conventional DTI for microstructural features, alternative multicompartiment diffusion models may offer increased biological specificity^{39,40}. Finally, we caution that these WM brain charts are not immediately suitable for the quantitative diagnosis of individual patients in a clinical setting. Substantial caveats exist even for the diagnostic interpretation of traditional anthropometric growth charts⁴¹. Similarly, considerable future research and validation will be necessary to translate these WM charts from a foundational research framework into a clinically validated diagnostic tool.

In conclusion, we have created comprehensive WM brain charts that define normative microstructural and macrostructural properties across the human lifespan, and have demonstrated their utility in identifying developmental milestones, detecting abnormalities in clinical populations and assessing relationships with cognitive and clinical variables. By openly sharing the underlying trajectories, scoring tools and harmonization pipelines, we aim to accelerate research into the structural basis of cognition, disease and ageing. This work lays the foundation for a new class of precision neuroscience tools grounded in normative WM architecture. The WM brain charts and code for aligning OOS datasets have been made available for researchers at <https://zenodo.org/records/15367425> (ref. 42).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10454-2>.

- Mesulam, M.-M. Large-scale neurocognitive networks and distributed processing for attention, language, and memory. *Ann. Neurol.* **28**, 597–613 (1990).
- Schmahmann, J. D. & Pandya, D. N. *Fiber Pathways of the Brain* (Oxford Univ. Press, 2006).
- Geschwind, N. Disconnection syndromes in animals and man. I. *Brain* **88**, 237–294 (1965).
- Catani, M. & Ffytche, D. H. The rises and falls of disconnection syndromes. *Brain* **128**, 2224–2239 (2005).
- Cole, T. J. The development of growth references and growth charts. *Ann. Hum. Biol.* **39**, 382–394 (2012).
- Bethlehem, R. A. I. et al. Brain charts for the human lifespan. *Nature* **604**, 525–533 (2022).
- Muller, A. O. et al. Advancing brain health: harnessing the centilebrain model for enhanced diagnosis of mental health disorders. *Ann. Med. Surg.* **86**, 4944–4946 (2024).
- Ge, R. et al. Normative modelling of brain morphometry across the lifespan with CentileBrain: algorithm benchmarking and model optimisation. *Lancet Digit. Health* **6**, e211–221 (2024).
- Habes, M. et al. The Brain Chart of Aging: machine-learning analytics reveals links between brain aging, white matter disease, amyloid burden, and cognition in the iSTAGING consortium of 10,216 harmonized MR scans. *Alzheimer Dementia* **17**, 89–102 (2021).
- Schilling, K. G. et al. Challenges in diffusion MRI tractography—lessons learned from international benchmark competitions. *Magn. Reson. Imaging* **57**, 194–209 (2019).
- World Health Organization. *WHO Child Growth Standards—Length/Height-for-Age, Weight-for-Age, Weight-for-Length, Weight-for-Height and Body Mass Index-for Age: Methods and Development* (2006).
- Grummer-Strawn, L. M., Reinold, C. M., Krebs, N. F. & Centers for Disease Control and Prevention (CDC). Use of World Health Organization and CDC growth charts for children aged 0–59 months in the United States. *MMWR Recomm. Rep.* **59** (RR9), 1–15 (2010).
- Beaulieu, C. The basis of anisotropic water diffusion in the nervous system — a technical review. *NMR Biomed.* **15**, 435–455 (2002).
- Yeh, F.-C. Shape analysis of the human association pathways. *NeuroImage* **223**, 117329 (2020).
- Rigby, R. A., Stasinopoulos, D. M. & Lane, P. W. Generalized additive models for location, scale and shape. *J. R. Stat. Soc. C* **54**, 507–554 (2005).
- Borgh, E. et al. Construction of the World Health Organization child growth standards: selection of methods for attained growth curves. *Stat Med* **25**, 247–265 (2006).
- Yeatman, J. D., Wandell, B. A. & Mezer, A. A. Lifespan maturation and degeneration of human brain white matter. *Nat. Commun.* **5**, 4932 (2014).
- Yakovlev, P. L. & Lecours, A.-R. in *Regional Development of the Brain in Early Life* (ed. Minkowski, A.) 3–70 (Blackwell, 1967).
- Benes, F. M. Myelination of cortical-hippocampal relays during late adolescence. *Schizophr. Bull.* **15**, 585–593 (1989).
- Forkel, S. J., Friedrich, P., Thiebaut de Schotten, M. & Howells, H. White matter variability, cognition, and disorders: a systematic review. *Brain. Struct. Funct.* **227**, 529 (2021).
- Pajevic, S. & Pierpaoli, C. Color schemes to represent the orientation of anisotropic tissues from diffusion tensor data: application to white matter fiber tract mapping in the human brain. *Magn. Reson. Med.* **42**, 526–540 (1999).
- Solmi, M. et al. Age at onset of mental disorders worldwide: large-scale meta-analysis of 192 epidemiological studies. *Mol. Psychiatry* **27**, 281–295 (2022).
- Li, M. et al. Integrative functional genomic analysis of human brain development and neuropsychiatric risks. *Science* **2018**, 362 (1979).
- Jones, D. K. *Diffusion MRI: Theory, Methods, and Application* (Oxford Univ. Press, 2010).
- de Faria, O. et al. Periods of synchronized myelin changes shape brain function and plasticity. *Nat. Neurosci.* **24**, 1508–1521 (2021).
- Flechsig, P. Developmental (myelogenetic) localisation of the cerebral cortex in the human subject. *Lancet* **158**, 1027–1030 (1901).
- Schilling, K. G. et al. White matter tract microstructure, macrostructure, and associated cortical gray matter morphology across the lifespan. *Imaging Neurosci.* **1**, imag1-00050 (2023).
- Lebel, C. et al. Diffusion tensor imaging of white matter tract evolution over the lifespan. *Neuroimage* **60**, 340–352 (2012).
- Grydeland, H. et al. Waves of maturation and senescence in micro-structural MRI markers of human cortical myelination over the lifespan. *Cereb. Cortex* **29**, 1369–1381 (2019).
- Reisberg, B. et al. Retrogenesis: clinical, physiologic, and pathologic mechanisms in brain aging, Alzheimer’s and other dementing processes. *Eur. Arch. Psychiatry Clin. Neurosci.* **249**, 3 (1999).
- Slater, D. A. et al. Evolution of white matter tract microstructure across the life span. *Hum. Brain Mapp.* **40**, 2252–2268 (2019).
- Liu, Y. et al. Connectivity-based topographical changes of the corpus callosum during aging. *Front. Aging Neurosci.* **13**, 753236 (2021).
- Villalón-Reina, J. E. et al. Lifespan normative modeling of brain microstructure. Preprint at *BioRxiv* <https://doi.org/10.1101/2024.12.15.628527> (2024).
- Zhu, A. H. et al. Lifespan reference curves for harmonizing multi-site regional brain white matter metrics from diffusion MRI. *Sci. Data* **12**, 748 (2025).
- Davis, S. W. et al. Assessing the effects of age on long white matter tracts using diffusion tensor tractography. *Neuroimage* **46**, 530–541 (2009).
- Di Biase, M. A. et al. Mapping human brain charts cross-sectionally and longitudinally. *Proc. Natl Acad. Sci. USA* **120**, e2216798120 (2023).
- Jbabdi, S. & Johansen-Berg, H. Tractography: where do we go from here. *Brain Connect* **1**, 169–183 (2011).
- Schilling, K. G. et al. Tractography dissection variability: what happens when 42 groups dissect 14 white matter bundles on the same dataset. *Neuroimage* **243**, 118502 (2021).
- Zhang, H., Schneider, T., Wheeler-Kingshott, C. A. & Alexander, D. C. NODDI: practical in vivo neurite orientation dispersion and density imaging of the human brain. *Neuroimage* **61**, 1000–1016 (2012).
- Novikov, D. S., Fieremans, E., Jespersen, S. N. & Kiselev, V. G. Quantifying brain microstructure with diffusion MRI: Theory and parameter estimation. *NMR Biomed.* **32**, e3998 (2019).
- Hendrickson, M. A. & Pitt, M. B. Three areas where our growth chart conversations fall short-room to grow. *JAMA Pediatr.* **176**, 123–124 (2022).

42. Kim, M. & Schilling, K. White matter microstructure and macrostructure brain charts across the human lifespan - models (0.2.1). *Zenodo* <https://doi.org/10.5281/zenodo.18435695> (2025).
43. Mukherjee, S. et al. Cognitive domain harmonization and cocalibration in studies of older adults. *Neuropsychology* **37**, 409–423 (2022).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2026, modified publication 2026

The HABS-HD Study Team

Sid O'Bryant²⁰,

²⁰Department of Family Medicine, University of North Texas Health Science Center, Fort Worth, TX, USA.

Alzheimer's Disease Neuroimaging Initiative **Arthur Toga**²¹

²¹Laboratory of Neuro Imaging, Department of Neurology, University of California School of Medicine, Los Angeles, CA, USA.

The BIOCARD Study Team **Marilyn Albert**²²

²²Department of Neurology, Johns Hopkins University, Baltimore, MD, USA.

Methods

Ethics

The research consists of secondary analyses of de-identified primary datasets. Information regarding informed consent of participants (or guardians) in primary studies can be found in the references in Supplementary Table S2.T2. Secondary analysis of these data was also approved by the Vanderbilt University Medical School Institutional Review Board (IRB no. 210968).

Software reporting

All analysis visualizations and statistics were generated using Python v3.9.20. Notable Python libraries used are pandas (v2.2.3), numpy (v2.0.2), seaborn (v0.13.2), matplotlib (v3.9.2), rpy2 (v3.5.11) and scipy (v1.13.1). For creating 3D qualitative visualizations of WM tracts, we used the mrview tool inside the MRtrix3 (version 3.0.4). Alternatively, for qualitative region of interest 3D visualizations, the OpenDIVE visualization library (v0.3.0) was used with a different Python (v3.10.18) for library compatibility purposes. Regarding the image preprocessing and processing software, please refer to ‘MRI processing pipeline’.

Data

We aggregated diffusion-weighted imaging (DWI) and T1-weighted (T1w) data from 50 independent studies spanning 0 to 100 years of age (Extended Data Table 1), encompassing 75,036 DWI scans from cognitively normal and clinical participants. All data were converted from DICOM to NIfTI using dcm2nii and organized in BIDS format in accordance with previous work⁴⁴. Notably, the 50 included datasets were harmonized through standardized preprocessing, postprocessing, and feature-wise statistical harmonization using GAMLSS.

MRI processing pipeline

DWI data were preprocessed using the PreQual pipeline⁴⁵, which corrects for susceptibility-induced, motion-related and eddy current distortions, and performs slice-wise signal imputation. Specifically, image denoising was performed using the MRtrix3 toolkit⁴⁶ (v3.0.4) implementation of Marchenko–Pastur principal component analysis. Following this, TOPUP from the FSL (v6.0.4) software library⁴⁷ is used for susceptibility-distortion correction. For DWI without reverse phase encoding scans acquired, TOPUP is run using a synthetic b0 image created from a T1w scan from the same imaging session via the Synb0-DISCO⁴⁸ algorithm (v3.1). FSL’s EDDY is then used for motion and eddy current-distortion correction, also performing slice-wise signal imputation with the flag ‘--repol’. Following preprocessing, DTI models were fit to all volumes with b-values $\leq 1,500 \text{ s mm}^{-2}$ (refs. 40,49) using dwi2tensor from MRtrix3. DTI-derived scalar maps—including FA, MD, AD and RD—were computed using tensor2metric from MRtrix3⁴⁶.

To enable consistent tract segmentation, all diffusion data were resampled to 1 mm isotropic resolution using MRtrix3⁵⁰. Tractography was performed using TractSeg⁵¹ (v2.8), which automatically segments 72 anatomically defined cerebral WM pathways (see Extended Data Table 2 for a list of tracts and abbreviations). For each tract, we computed streamline density-weighted averages of DTI metrics (FA, MD, AD and RD) as well as macrostructural features—volume, length and surface area—using the Scilpy toolkit⁵² (v1.5.0) scripts scil_compute_bundle_mean_std.py and scil_evaluate_bundles_individual_measures.py, respectively. Both raw and ratio-normalized macrostructural features are provided, where volume, surface area and average length of tracts are normalized to total brain volume (excluding ventricles), estimated total intracranial volume and cerebral WM volume.

T1w images were included only when acquired in the same session as DWI data. Brain segmentation was performed using the recon-all command from FreeSurfer (v7.2.0)⁵³, yielding estimates of cerebral WM volume, brain volume excluding ventricles and total intracranial volume. For participants aged ≤ 2 years, we employed infant_recon_all

from the infant FreeSurfer pipeline⁵⁴ (v0.0) to account for age-specific brain morphology. Cerebral WM masks were defined using MRtrix3⁴⁶ STT labels, excluding the cerebellum and brainstem. T1w images and corresponding WM segmentations were rigidly registered to DWI space using FSL’s⁴⁷ epi_reg. Whole-brain WM DTI metrics were then computed by averaging values within the WM mask.

The PreQual preprocessing pipeline can be downloaded from <https://zenodo.org/records/14058394>. We also make the entire postprocessing pipeline available at <https://zenodo.org/records/17144460>.

Data selection

Following the approach of Bethlehem et al.⁶, we restricted our analysis to cross-sectional data. Participant scans were excluded if key demographic information—age, sex or cognitive status—was missing or not reported for that participant. For each dataset, sex was used as reported and encoded as a binary variable, whereas age was a continuous value in years. Regarding infants in the BABIES-ABC, HBCD and dHCP datasets, we used corrected age based on the due date to better account for prematurity and provide a more accurate developmental context.

Quality control (QC) procedures followed our previously established framework for large-scale dMRI analysis⁵⁵. QC metrics were visually reviewed across PreQual preprocessing outputs, FreeSurfer segmentations, and T1w–DWI registration results. Scans were excluded if any preprocessing step failed or was deemed unusable. TractSeg outputs were excluded if more than 12 of the 72 tracts failed to reconstruct. Outlier removal was performed separately for each tract-level feature within each dataset, with observations excluded if exceeding ± 4 standard deviations from the mean. We note it is possible for FreeSurfer to have failed with successful tractography, and vice-versa, in which cases we retained only data from the contrast that passed quality assessment. The number of sessions and participants retained after QC is detailed in Extended Data Table 1.

Normative trajectories were modelled using only participants labelled as ‘typically developing’ or ‘control’ within their respective studies to reflect patterns of healthy WM development and ageing. Participants who were initially labelled as typically ageing but later transitioned to a clinical diagnosis were also excluded when fitting normative trajectories. At the tract level, features were retained only if the tract was reconstructed with the full target of 2,000 streamlines—the default setting in TractSeg.

Normative modelling of features

We utilized GAMLSS¹⁵, a flexible regression framework endorsed by the World Health Organization for constructing normative growth curves^{12,16}. GAMLSS extends generalized linear and additive models by allowing the simultaneous estimation of multiple distribution parameters—not only the mean, but also variance, skewness and kurtosis—through functions that vary with age and other covariates. This flexibility enables precise modelling of lifespan trajectories for WM features.

Formally, GAMLSS defines each parameter of the assumed response distribution through additive predictors:

$$g_k(\theta_k) = X_k \beta_k + \sum_{j=1}^{J_k} Z_{kj} \gamma_{kj}, \quad (1)$$

where $g_k(\cdot)$ is the link function for the k th parameter, X_k and β_k are the design matrix and fixed effects. The summation incorporates J_k smooth functions, where Z_{kj} are the design matrices for the basis expansions and γ_{kj} are the corresponding smoothing coefficients. This formulation allows flexible, non-linear modelling of the entire distribution as a function of covariates such as age, sex and dataset.

Previous work has shown dMRI-derived features to have skewed distributions. Following Bethlehem et al.⁶, we used the generalized gamma (GG) distribution with fractional polynomial (fp) fitting to

estimate lifespan trajectories. The GG distribution offers substantial flexibility, accommodating a broad range of distributional shapes, and is therefore suitable for modelling both microstructural and macrostructural imaging features. The model parameters, location (μ), scale (σ) and shape (v), were defined as:

$$g_1(\mu) = x_{\text{sex}}\beta_{1,\text{sex}} + \text{fp}_1(x_{\text{age}}) + \gamma_{1,D} \quad (2)$$

$$g_2(\sigma) = x_{\text{sex}}\beta_{2,\text{sex}} + \text{fp}_2(x_{\text{age}}) + \gamma_{2,D} \quad (3)$$

$$g_3(v) = \alpha, \quad (4)$$

where $g_1(\cdot)$ and $g_2(\cdot)$ denote log link functions, whereas $g_3(\cdot)$ is an identity link. Age was modelled as a continuous predictor using fractional polynomial transformations $\text{fp}_k(\cdot)$, enabling non-linear representation of age-related effects. Sex was included as a binary fixed effect (x_{sex}) and dataset-specific variability was modelled via random intercepts ($\gamma_{1,D}$ and $\gamma_{2,D}$). The GG distribution was parameterized as described in Rigby et al.⁵⁶ to enable compatibility with GAMLSS framework. Model selection was performed by the Bayesian information criterion⁵⁷ across all combinations of 1–3 fractional polynomial terms for the μ and σ parameters. The shape parameter v was treated as a global offset without age- or dataset-specific effects, consistent with Bethlehem et al.⁶.

We modelled normative trajectories for 509 features, including 288 tract-level (72 tracts $\times$ 4 features) microstructural measures (mean FA, MD, AD and RD) and 216 macrostructural (72 tracts $\times$ 3 features) measures (volume, average length and surface area). An additional five global WM features—mean FA, MD, AD, RD and WM volume—were also modelled. Macrostructural features were further normalized using total intracranial volume, brain volume excluding ventricles, and global WM volume to account for anatomical scaling.

To derive scaling factors for normalization, we assumed a spherical brain geometry and estimated radius and surface area using:

$$r_{\text{Brain}} = \left(\frac{3V_{\text{Brain}}}{4\pi} \right)^{\frac{1}{3}} \quad (5)$$

and

$$\text{SA}_{\text{Brain}} = 4\pi r_{\text{Brain}}^2 \quad (6)$$

respectively, where V_{Brain} denotes the normalizing volume, with separate scaling factors derived for estimated total intracranial volume, brain volume excluding ventricles, and global WM volume. V_{Brain} , SA_{Brain} and r_{Brain} (Equations 5 and 6) were used to normalize tract-level volume, surface area and average length metrics, respectively.

When considering both normalized and unnormalized features, we had 1,157 features in total for which we created lifespan curves: 509 unnormalized (288 microstructure, 216 macrostructure and 5 global) and 648 normalized (216 macrostructure normalized in 3 separate ways) features.

WM lifespan milestones

For assessing myelination, we studied the median RD trajectories of tract-specific brain charts. Following the work from Yeatman et al.¹⁷, we performed piecewise segmentations of the trajectories to assess whether the myelination time points align with proposed periods of myelin development and maturation derived in refs. 18,25,26. Specifically, we performed a piecewise segmentation of the median trajectories to identify three break points that minimize the sum of squared errors across the resulting linear segments (where the piecewise model was fit between ages 1 and 90 years). The segments correspond to myelination periods of rapid development in infancy, slower myelination in adolescence, the stable plateau of young adulthood, and the decline

of ageing. For each segment, we calculated both the average slope and the per cent change per year (Supplementary Table SDT.T5).

Centile scores across cognitive groups

We used the fitted normative trajectories and dataset-specific random effects to compute individualized centile scores for non-control participants across all tract-level features. In total, 6,777 participants from 16 clinical groups passed QC and were included in the analysis (Extended Data Table 3).

As the analysis was restricted to cross-sectional data, we selected a single scan per participant. For individuals with longitudinal data, we retained the most recent scan corresponding to their most severe clinical diagnosis. For example, participants progressing from cognitively unimpaired to MCI and then to Alzheimer's disease were classified based on their most recent scan with a clinical Alzheimer's disease or dementia diagnosis.

For each clinical group, we compared the distribution of centile scores across tract metrics to the 50th percentile expected in the normative population. In addition, we quantified overall deviation from the normative distribution using the nCMD for each participant j (nCMD _{j}):

$$\text{nCMD}_j = \frac{\sqrt{(x - \mu)^T S^{-1} (x - \mu)}}{N_{j,\text{tot}}} \quad (7)$$

where $N_{j,\text{tot}}$ is the number of reconstructed tract features for participant j , x is a vector of centile scores (ranging from 0 to 1), $\mu = 0.5$ denotes the median centile, and S^{-1} is the inverse of the covariance matrix for the features.

We computed nCMD separately for tract-level microstructural features, macrostructural features, and their combination. Group-level comparisons were then made by evaluating the median nCMD in each non-control group relative to the distribution observed in the cognitively normal population, where statistical significance was tested using a one-sample Wilcoxon test (two-sided) after Bonferroni correction for multiple comparisons.

Alignment for OOS datasets

A central utility of normative brain charts is their use as reference models for external, OOS datasets. To enable this, new datasets must be aligned to the fitted trajectories by estimating study-specific offsets for the distributional parameters. Within the GAMLSS framework, each dataset D is modelled with random effect terms $\gamma_{1,D}$ and $\gamma_{2,D}$, which account for dataset-specific variabilities in the location (μ) and scale (σ) parameters, respectively. For an unseen dataset S , alignment involves estimating these two unknown offsets. We demonstrate this process using the ADNI dataset, where we fit additional models without ADNI for the purpose of demonstrating OOS alignment.

For a given brain chart modelling a tract-level metric, the GAMLSS-based lifespan distribution is defined as:

$$g_1(\mu) = \text{fp}_1(x_{\text{age}}) + x_{\text{sex}}\beta_{1,\text{sex}} + \gamma_{1,S} \quad (8)$$

$$g_2(\sigma) = \text{fp}_2(x_{\text{age}}) + x_{\text{sex}}\beta_{2,\text{sex}} + \gamma_{2,S} \quad (9)$$

$$g_3(v) = \alpha, \quad (10)$$

where $\gamma_{1,S}$ and $\gamma_{2,S}$ are dataset-specific random effects for the ADNI dataset, and all other model parameters are fixed from the trained GAMLSS model fitted without ADNI data.

As all other effects are known from the reference model, alignment requires estimating only the unknown random effects $\gamma_{1,S}$ and $\gamma_{2,S}$. We initialized both to zero and restricted the estimation procedure to typically ageing individuals in the ADNI cohort. To minimize diagnostic

Article

confounding, we selected only participants with no history or future diagnosis of MCI or Alzheimer's disease and used their earliest available scan. This cognitively normal subset is denoted S_{CN} .

For each cognitively normal participant j in S_{CN} , we computed the likelihood of their observed metric M_j under the GG distribution defined by equations (8)–(10). Each of the resulting probability densities $P_j = GG(M_j | \mu, \sigma, v)$ were used to calculate the overall negative log-likelihood across all participants in S_{CN} :

$$\text{Negative log-likelihood} = - \sum_{j=1}^{|S_{CN}|} \log(P_j), \quad (11)$$

We iteratively re-estimated $\gamma_{L,S}$ and $\gamma_{2,S}$ by minimizing the negative log-likelihood until convergence. The final estimates, $\hat{\gamma}_{L,S}$ and $\hat{\gamma}_{2,S}$, were then applied to all participants in the ADNI dataset to compute adjusted centile scores along to the normative reference space.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All derived data and associated demographic information from publicly available datasets, as well as from datasets for which the Data Use Agreement (DUA) permits sharing, have been made accessible at <https://zenodo.org/records/18891847> (ref. 58). For datasets where the DUA does not permit redistribution of derivative data, these materials have been returned to or shared directly with the respective data custodians. Data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) are available upon request from <https://adni.loni.usc.edu/>. Data from the AOMIC-PIOP1 dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds002785/versions/2.0.0>. Data from the AOMIC-PIOP2 dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds002790/versions/2.0.0>. Data from the AOMIC-ID1000 dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds003097/versions/1.2.1>. Data from the Boston Adolescent Neuroimaging of Depression and Anxiety (BANDA) dataset are available upon request from <https://www.humanconnectome.org/study/connectomes-related-anxiety-depression>. Data from BIOCARD are available upon request after filling out a data use application: <https://www.gaaindata.org/partner/BIOCARD>. Data from the Baltimore Longitudinal Study of Aging (BLSA) are available upon request from <https://www.blsa.nih.gov/>. Data from the Calgary Preschool MRI Dataset are freely available for download at: <https://doi.org/10.17605/OSF.IO/AXZ5R>. Data from the Centre for Attention Learning and Memory (CALM) dataset are available upon request from <https://calm.mrc-cbu.cam.ac.uk/researchers/>. Data from the Cambridge Center for Ageing Neuroscience (CAMCAN) dataset are available upon request from <https://camcan-archive.mrc-cbu.cam.ac.uk/dataaccess/>. Data from the Dallas Lifespan Brain Study (DLBS) dataset are freely available for download on OpenNeuro at <https://openneuro.org/datasets/ds004856/versions/1.2.0>. Data from the Health and Aging Brain Study—Health Disparities (HABS-HD) dataset are available upon request from <https://apps.unthsc.edu/itr/reports>. Imaging data and basic demographic information for the Healthy Brain Network (HBN) dataset are freely available to download from https://fcon_1000.projects.nitrc.org/indi/cmi_healthy_brain_network/. Phenotypic data are available upon request by filling out a data use agreement (https://fcon_1000.projects.nitrc.org/indi/cmi_healthy_brain_network/Phenotypic.html). Data from the Human Connectome Project—Aging (HCPA) dataset are available upon request from <https://www.humanconnectome.org/study/hcp-lifespan-aging>. Data from the Lifespan Baby Connectome Project (HCPBaby) dataset are

available upon request from <https://www.humanconnectome.org/study/lifespan-baby-connectome-project>. Data from the Human Connectome Project—Development (HCPD) dataset are available upon request from <https://www.humanconnectome.org/study/hcp-lifespan-development>. Data from the Human Connectome Project—Young Adult (HCP) dataset are freely available for download from <https://www.humanconnectome.org/study/hcp-young-adult>. Data from the Infant Brain Imaging Study (IBIS) are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=19. Data from the International Consortium for Brain Mapping (ICBM) dataset are available upon request from www.loni.usc.edu/ICBM. Data from the Longitudinal Brain Correlates of Multisensory Lexical Processing in Children study (shortened to Lexical in this manuscript) are freely available for download on OpenNeuro (<https://openneuro.org/datasets/ds001894/versions/1.4.2>). Data from the Memory and Aging Project (MAP), Religious Orders Study (ROS) and the Minority Aging Research Study (MARS) datasets are available upon request from <https://www.radc.rush.edu/>. More information about participant demographics and study information can be found at <https://www.rushu.rush.edu/research-rush-university/departamental-research/rush-alzheimers-disease-center/rush-alzheimers-disease-center-research/epidemiologic-research>. Data from the Multisite, Multiscanner, and Multisubject Acquisitions for Studying Variability in Diffusion Weighted Magnetic Resonance Imaging (MASiVar) dataset are freely available for download on OpenNeuro at <https://openneuro.org/datasets/ds003416/versions/2.0.2>. Data from the National Alzheimer's Coordinating Center (NACC) and the Standardized Centralized Alzheimer's and Related Dementias Neuroimaging (SCAN) are available upon request from <https://naccdata.org/requesting-data/data-request-process>. Imaging data and basic demographic information for the Nathan Kline Institute—Rockland Sample (NKI) dataset are freely available to download from <https://rocklandsample.org/accessing-the-neuroimaging-data-releases>. Phenotypic data are available upon request by filling out a data use agreement (<https://rocklandsample.org/phenotypic-data>). Data from the Pediatric Imaging, Neurocognition, and Genetics dataset (PING) are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=2607. Data from the Queensland Twin Adolescent Brain (QTAB) dataset are freely available for download on OpenNeuro at <https://openneuro.org/datasets/ds004146/versions/1.0.4>. Data from the UCLA Consortium for Neuropsychiatric Phenomics LA5c Study (UCLA) dataset are freely available for download on OpenNeuro at <https://openneuro.org/datasets/ds000030/versions/1.0.0>. Data coming from the Southwestern University (SWU) dataset, part of the Consortium for Reliability and Reproducibility (CoRR), were downloaded via NITRC-IR from the 1000 Functional Connectomes Project. In order to access the CoRR datasets through NITRC, users must be logged into NITRC at the time of download and registered with the 1000 Functional Connectomes Project/INDI website. More information about this subset can be found at https://fcon_1000.projects.nitrc.org/indi/CoRR/html/swu_4.html. Data from the Southwest University (SWU) Longitudinal Imaging Multimodal dataset are freely available for download at https://fcon_1000.projects.nitrc.org/indi/retro/southwestuni_qiu_index.html. Data from the Social Reward and Nonsocial Reward Processing Across the Adult Lifespan: An Interim Multi-echo fMRI and Diffusion Dataset (referred to as TempleSocial in this manuscript) are freely available for download on OpenNeuro at <https://openneuro.org/datasets/ds005123/versions/1.1.3>. Data from UK Biobank (UKBB) are available upon request from <https://www.ukbiobank.ac.uk/>. Data from the UPennRisk dataset are freely available for download on OpenNeuro at <https://openneuro.org/datasets/ds002843/versions/1.0.1>. Data from the dataset 'A longitudinal neuroimaging dataset on language processing in children ages 5, 7, and 9 years old' (referred to as UTAustin579 in

this manuscript) are freely available for download on OpenNeuro at <https://openneuro.org/datasets/ds003604/versions/1.0.7>. Data from the Vanderbilt Memory and Aging Project (VMAP JEFFERSON, VMAP_2.0, TN Aging Project) are available for download upon request from <https://vmacdata.org/vmap/data-requests>. Data from the Wisconsin Registry for Alzheimer's Prevention (WRAP) are available upon request from <https://wrap.wisc.edu/data-requests-2/>. Data from the HEALTHy Brain and Child Development (HBCD) Study are available for download upon request from <https://hbcdstudy.org/data-sharing/>. Data from the Adolescent Brain Cognitive Development (ABCD) Study are available for download upon request from (<https://abcdstudy.org/scientists/data-sharing/>). Data used in this manuscript from the Age-ility Project (Phase I only) are available for download from <https://www.nitrc.org/projects/age-ility>. Data from the Early Brain Development in Twins (EBDT) dataset are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=2384. Data from the Bipolar and Schizophrenia Consortium for Parsing Intermediate Phenotypes dataset (BSNIP1) and its renewal (BSNIP2) are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=2274 and https://nda.nih.gov/edit_collection.html?id=2165. Data from the Developing Human Connectome Project (dHCP) are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=3955. Vanderbilt University data (MORGAN, BABIES-ABC, CUTTING, VUMC-ASD) are subject to third party restrictions. Please contact corresponding authors for data requests.

Code availability

Code for OOS data alignment, obtaining centile curves, and fitting the GAMLSS models is available in a containerized Docker image that is downloadable from <https://zenodo.org/records/15367425> (ref. 42). Instructions for running the Docker image can also be found at this URL and in the Supplementary Information. The postprocessing pipeline to obtain the microstructural and macrostructural measurements from dMRI data is also available as a Docker image downloadable from https://hub.docker.com/r/kimm58/wm_lifespan_processing. Instructions for running the Docker image are available at <https://zenodo.org/records/17144461> (ref. 59).

44. Kim ME, et al. Scalable, reproducible, and cost-effective processing of large-scale medical imaging datasets. In *Medical Imaging 2025: Imaging Informatics* (ed. Wu, S.) **13411**, 128–140 (SPIE, 2025).
45. Cai, L. Y. et al. PreQual: an automated pipeline for integrated preprocessing and quality assurance of diffusion weighted MRI images. *Magn. Reson. Med.* **86**, 456–470 (2021).
46. Tournier, J. D. et al. MRtrix3: a fast, flexible and open software framework for medical image processing and visualisation. *Neuroimage* **202**, 116137 (2019).
47. Jenkinson, M., Beckmann, C. F., Behrens, T. E. J., Woolrich, M. W. & Smith, S. M. FSL. *Neuroimage* **62**, 782–90 (2012).
48. Schilling, K. G. et al. Synthesized b0 for diffusion distortion correction (Synb0-DisCo). *Magn. Reson. Imaging* **64**, 62–70 (2019).
49. Tang, L., Zhou, X. J. & Diffusion, M. R. I. of cancer: from low to high b-values. *J. Magn. Reson. Imaging* **49**, 23–40 (2019).
50. Theaud, G. et al. TractoFlow: a robust, efficient and reproducible diffusion MRI pipeline leveraging Nextflow & Singularity. *Neuroimage* **218**, 116889 (2020).
51. Wasserthal, J., Neher, P. & Maier-Hein, K. H. TractSeq - fast and accurate white matter tract segmentation. *Neuroimage* **183**, 239–253 (2018).
52. Renauld, E. et al. Tractography analysis with the scilpy toolbox. *Aperture Neuro* <https://doi.org/10.52294/001c.154022> (2026).
53. Fischl, B. FreeSurfer. *Neuroimage* **62**, 774–781 (2012).
54. Zöllei, L., Iglesias, J. E., Ou, Y., Grant, P. E. & Fischl, B. Infant FreeSurfer: an automated segmentation and surface extraction pipeline for T1-weighted neuroimaging data of infants 0–2 years. *Neuroimage* **218**, 116946 (2020).
55. Kim, M. E. et al. Scalable quality control on processing of large diffusion-weighted and structural magnetic resonance imaging datasets. *PLoS ONE* **20**, e0327388 (2025).
56. Rigby, R., Stasinopoulos, M., Heller, G. & De Bastiani, F. *Distributions for Modelling Location, Scale and Shape: Using GAMLSS in R* (CRC Press, 2017).
57. Neath, A. A. & Cavanaugh, J. E. The Bayesian information criterion: background, derivation, and applications. *Wiley Interdiscip. Rev. Comput. Stat.* **4**, 199–203 (2012).
58. Kim, M., Landman, B. & Schilling, K. Data for white matter micro and macrostructure brain charts for the human lifespan. *Zenodo* <https://doi.org/10.5281/zenodo.18891848> (2026).

59. Kim, M. White matter microstructure and macrostructure brain charts across the human lifespan - post-processing container (0.1). *Zenodo* <https://doi.org/10.5281/zenodo.17144461> (2025).
60. Holmes, J., Bryant, A., The CALM Team & Gathercole, S. E. Protocol for a transdiagnostic study of children with problems of attention, learning and memory (CALM). *BMC Pediatr.* **19**, 10 (2019).

Acknowledgements This work was supported in part by the National Institute of Health through NIH awards K01-EB032898 (K.G.S.) and K01-AG073584 (D.A.), grant number 1R01EB017230-01A1 (B.A.L.) and K24-AG046373 (A.L.J.) and ViSE/VICTR VR3029, UL1-TR000445 and UL1-TR002243. This work was supported by NIA grants R01-AG034962 (A.L.J.), R01-AG056534 (A.L.J.) and R01-AG062826 and Alzheimer's Association IIRG-08-88733 (A.L.J.) and the NICHD, R01-HD114489 (S.V.-B.). This work was supported by the Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium (ADSP-PHC), which is funded by NIA (U24 AG074855, U01 AG068057 and R01 AG059716). This work was conducted in part using the resources of the Advanced Computing Center for Research and Education (ACCRE) at Vanderbilt University, Nashville, TN. We appreciate the National Institute of HealthS10 Shared Instrumentation grant 1S10OD020154-01, and grant 1S10OD023680-01 (Vanderbilt's High-Performance Computer Cluster for Biomedical Research). This work was also supported in part by Intramural Research Program of the National Institute on Aging, NIH. BABIES and ABC datasets were supported by the Jacobs Foundation Early Career Research Fellowship (2017-1261-05) (K.L.H.); National Science Foundation CAREER Award (2042285) (K.L.H.); Brain and Behavior Research Foundation John and Polly Sparks Foundation Investigator Award (29593) (K.L.H.); Vanderbilt Institute for Clinical and Translational Research Grant (VR53419) (K.L.H.); Vanderbilt Strong Grant; Vanderbilt Kennedy Center Grant (K.L.H.); and National Institute of Mental Health (R01MH129634) (K.L.H.). This work was supported by the NIH NINDS through award numbers R01 NS108445 and R01 NS10130 (V.L.M.), and R01 NS134625 and R01 NS112252 (D.J.E.). This work was supported by the NIH Eunice Kennedy Shriver National Institute Of Child Health and Human Development through grant number F31 HD104385 (T.Q.N.), P50HD103537 (J. L. Neul), R37 HD095519 MERIT Award, R01 HD089474, R01 HD109151, R01 HD067254 and R01 HD044073 (L.E.C.). This work was also supported by the NIH NINDS through grant number R01 NS049096 (L.E.C.). Research reported in this publication was supported by NIGMS of the National Institutes of Health under award number T32GM007347 and T32GM152284. Data used in preparation of this article were derived from BIOCARD study, supported by grant U19 – AG033655 from the National Institute on Aging. The BIOCARD study team did not participate in the analysis or writing of this report, however, they contributed to the design and implementation of the study. A listing of BIOCARD investigators can be found on the BIOCARD website (on the 'BIOCARD Data Access Procedures' page, 'Acknowledgement Agreement' document), or at <https://biocard.pathology.jhu.edu/our-team/>. Data and/or research tools used in the preparation of this manuscript were obtained from the National Institute of Mental Health (NIMH) Data Archive (NDA). NDA is a collaborative informatics system created by the National Institutes of Health to provide a national resource to support and accelerate research in mental health. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or of the submitters submitting original data to NDA. Data were provided by the Human Connectome Project, WU-Minn Consortium (principal investigators: David Van Essen and Kamil Ugurbil; 1U54MH091657) funded by the 16 NIH Institutes and Centers that support the NIH Blueprint for Neuroscience Research; and by the McDonnell Center for Systems Neuroscience at Washington University. VMAP data were collected by Vanderbilt Memory and Alzheimer's Center Investigators at Vanderbilt University Medical Center. VMAP began in 2012 with the goal of investigating vascular health and brain aging. BLSA is a prospective cohort study with continuous enrollment that began in 1958. Comprehensive data from BLSA are available upon request by a proposal submission through the cohort website (<https://www.blsa.nih.gov>). The BLSA is supported by the Intramural Research Program of the National Institute on Aging, NIH. Data collection and sharing for this project was provided by the Cambridge Centre for Ageing and Neuroscience (CamCAN). CamCAN funding was provided by the UK Biotechnology and Biological Sciences Research Council (grant number BB/H008217/1), together with support from the UK Medical Research Council and University of Cambridge, UK. Data used in the preparation of this work were obtained from the Cambridge Centre for Ageing and Neuroscience (CamCAN) repository (<http://www.mrc-cbu.cam.ac.uk/datasets/camcan/>). Cam-CAN is a large-scale collaborative research project at the University of Cambridge. The investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf. Data collection and sharing for ADNI were supported by National Institutes of Health Grant U01-AG024904 and Department of Defense (award number W81XWH-12-2-0012). ADNI is also funded by the National Institute on Aging, the National Institute of Biomedical Imaging and Bioengineering, and through generous contributions from the following: AbbVie, Alzheimer's Association; Alzheimer's Drug Discovery Foundation; Araclon Biotech; BioClinica; Biogen; Bristol-Myers Squibb Company; CereSpir; Cogstate; Eisai; Elan Pharmaceuticals; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd and its affiliated company Genentech; Fujirebio; GE Healthcare; IXICO; Janssen Alzheimer Immunotherapy Research & Development; Johnson & Johnson Pharmaceutical Research & Development; Lumosity; Lundbeck; Merck & Co.; Meso Scale Diagnostics; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics. The Canadian Institutes of Health Research is providing funds to support ADNI clinical sites in Canada. Private sector contributions are facilitated by the Foundation for the National Institutes of Health (<https://www.fnih.org>). The grantee organization is the Northern California Institute for Research and Education, and the study is coordinated by the Alzheimer's Therapeutic Research Institute at the University of Southern California. ADNI data are disseminated by the Laboratory for Neuro Imaging at the University of Southern California. Research reported in this publication as a part of the Health and Aging Brain Study: Health Disparities (HABS-HD) was supported by the National Institute on Aging of the National Institutes of Health under award numbers R01AG054073 and R01AG058533, R01AG070862, P41EB015922 and U19AG078109. The content is solely the responsibility of the authors and does not necessarily represent the official views of the

Article

National Institutes of Health. The NACC database is funded by NIA/NIH grant U24 AG072122. NACC data are contributed by the NIA-funded ADCRCs: P30 AG062429 (J. Brewer), P30 AG066468 (O. Lopez), P30 AG062421 (B. Hyman), P30 AG066509 (T. Grabowski), P30 AG066514 (M. Sano), P30 AG066530 (H. Chui), P30 AG066507 (M. Albert), P30 AG066444 (J. Morris), P30 AG066518 (J. Kaye), P30 AG066512 (T. Wisniewski), P30 AG066462 (S. Small), P30 AG072979 (D. Wolk), P30 AG072972 (C. DeCarli), P30 AG072976 (A. Saykin), P30 AG072975 (D. Bennett), P30 AG072978 (A. McKee), P30 AG072977 (R. Vassar), P30 AG066519 (F. LaFerla), P30 AG062677 (R. Petersen), P30 AG072980 (E. Reiman), P30 AG062422 (G. Rabinovici), P30 AG066511 (A. Levey), P30 AG072946 (L. Van Eldik), P30 AG062715 (S. Asthana), P30 AG072973 (R. Swerdlow), P30 AG066506 (T. Golde), P30 AG066508 (S. Strittmatter), P30 AG066515 (V. Henderson), P30 AG072947 (S. Craft), P30 AG072931 (H. Paulson), P30 AG066546 (S. Seshadri), P20 AG068024 (E. Roberson), P20 AG068053 (J. Miller), P20 AG068077 (G. Rosenberg), P20 AG068082 (A.L.J.), P30 AG072958 (H. Whitson), P30 AG072959 (J. Leverenz). This research has been conducted using the UK Biobank resource, application 16315. Data contributed from MAP/ROS/MARS was supported by NIA R01AG017917, P30AG10161, P30AG072975, R01AG022018, R01AG056405, UH2NS100599, UH3NS100599, R01AG064233, R01AG15819 and R01AG067482, and the Illinois Department of Public Health (Alzheimer's Disease Research Fund). The data contributed from the Wisconsin Registry for Alzheimer's Prevention was supported by NIA AG021155, AG027161, AG037639, and AG054047. We used generative artificial intelligence (AI) to create code segments based on task descriptions, as well as debug, edit, and autocompile code. Additionally, generative AI technologies were used to assist in structuring sentences and performing grammatical checks. It is imperative to highlight that the conceptualization, ideation and all prompts provided to the AI originate entirely from the authors' creative and intellectual efforts. We take accountability for the review of all content generated by AI in this work. Data collection and sharing for this project was provided by the Centre for Attention, Learning and Memory (CALM). CALM funding was provided by the UK Medical Research Council and University of Cambridge, UK. Data used in the preparation of this work were obtained from CALM resource (<https://calm.mrc-cbu.cam.ac.uk/>). The study protocol is reported in ref. 60. Data from the VUMC-ASD dataset are collected seeking to elucidate the neural mechanisms underlying atypical sensory processing in autism spectrum disorders, focusing on tactile and interoceptive sensitivity and their relationship to social and behavioral symptoms. By integrating behavioral, neurophysiological and neuroimaging approaches, the studies aim to identify how alterations in thalamocortical and salience networks contribute to sensory hypo- and hyper-responsiveness in autism spectrum disorders, ultimately informing early identification and novel intervention strategies. Collection of these data was supported by K01MH090232, R21MH101321 and R01MH102272. The NACC database is funded by NIA/NIH Grant U24 AG072122. SCAN is a multi-institutional project that was funded as a U24 grant (AG067418) by the National Institute on Aging in May 2020. Data collected by SCAN and shared by NACC are contributed by the NIA-funded ADCRCs as follows: Arizona Alzheimer's Center—P30 AG072980 (E. Reiman); R01 AG069453 (E. Reiman); P30 AG019610 (E. Reiman); and the State of Arizona which provided additional funding supporting our centre; Boston University—P30 AG013846 (N. Kowall); Cleveland ADCRC—P30 AG062428 (J. Leverenz); Cleveland Clinic, Las Vegas—P20AG068053; Columbia—P50 AG008702 (S. Small); Duke/UNC ADCRC—P30 AG072958; Emory University—P30AG066511 (L. Allan); Indiana University—R01 AG19771 (A. Saykin); P30 AG10133 (A. Saykin); P30 AG072976 (A. Saykin); R01 AG061788 (S. Risacher); R01 AG053993 (Y.-C. Wu); U01 AG057195 (L. Apostolova); U19 AG063911 (B. Boeve); and the Indiana University Department of Radiology and Imaging Sciences; Johns Hopkins—P30 AG066507 (M. Albert); Mayo Clinic—P50 AG016574 (R. Petersen); Mount Sinai—P30 AG066514 (M. Sano); R01 AG054110 (T. Hedden); R01 AG053509 (T. Hedden); New York University—P30AG066512-01S2 (T. Wisniewski); R01AG056031 (R. Osorio, MD); R01AG056531 (R. Osorio and G. Jean-Louis); Northwestern University—P30 AG013854 (R. Vassar); R01 AG045571 (E. Rogalski); R56 AG045571 (E. Rogalski); R01 AG067781 (E. Rogalski); U19 AG073153 (E. Rogalski); R01 DC008552 (M.-M. Mesulam); R01 AG077444 (M.-M. Mesulam and E. Rogalski); R01 NS075075 (E. Rogalski); R01 AG056258 (E. Rogalski); Oregon Health and Science University—P30 AG008017 (J. Kaye); R56 AG074321 (J. Kaye); Rush University—P30 AG010161 (D. Bennett); Stanford—P30AG066515; P50 AG047366 (V. Henderson); University of Alabama, Birmingham—P20; University of California, Davis—P30 AG10129 (C. DeCarli); P30 AG072972 (C. DeCarli); University of California, Irvine—P50 AG016573 (F. LaFerla); University of California, San Diego—P30AG062429 (J. Brewer); University of California, San Francisco—P30 AG062422 (G. Rabinovici); University of Kansas—P30 AG035982 (R. Swerdlow); University of Kentucky—P30 AG028283-15S1 (L. Van Eldik and B. Gold); University of Michigan ADCRC—P30AG053760 (H. Paulson), P30AG072931 (H. Paulson), Cure Alzheimer's Fund 200775 (H. Paulson), U19 NS120384 (C. DeCarli and H. Paulson), R01 AG068338 (B. Giordani, C. Persad and Y. Murphey), S10OD026738-01 (D. Noll), R01 AG058724 (B. Hampstead), R35 AG072262 (B. Hampstead), W81XWH2110743 (B. Hampstead), R01 AG073235 (N. Chiaravallotti and B. Hampstead), 1I01RX001534 (B. Hampstead), IRX001381 (B. Hampstead); University of New Mexico—P20 AG068077 (G. Rosenberg); University of Pennsylvania—state of PA project 2019NF4100087335 (D. Wolk); Rooney Family Research Fund (D. Wolk); R01 AG055005 (D. Wolk); University of Pittsburgh—P50 AG005133 (O. Lopez); University of Southern California—P50 AG005142 (H. Chui); University of Washington—P50 AG005136 (T. Grabowski); University of Wisconsin—P50 AG033514 (S. Asthana); Vanderbilt University—P20 AG068082; Wake Forest—P30AG072947 (S. Craft); Washington University, St Louis—P01 AG03991 (J. Morris); P01 AG026276 (J. Morris); P20 MH071616 (D. Marcus); P30 AG066444 (J. Morris); P30 NS098577 (D. Marcus); R01 AG021910 (R. Buckner); R01 AG043434 (C. Roe); R01 EB009352 (D. Marcus); UL1 TR000448 (B. Evanoff); U24 RR021382 (B. Rosen); Avid Radiopharmaceuticals/Elis Lilly; Yale—P50 AG047270 (S. Strittmatter); R01AG052560 (C. van Dyck and R. Carson); R01AG062276 (C. van Dyck); tFlorida—P30AG066506-03 (G. Smith); P50 AG047266 (T. Golde). Data used in the preparation of this article were obtained from the HEALTHY Brain and Child Development (HBCD) Study (<https://hbcdstudy.org/>), held in the NIH Brain Development Cohorts Data Sharing Platform. The HBCD Study is supported by the NIH and additional federal partners under award numbers U01DA055352, U01DA055353, U01DA055366, U01DA055365, U01DA055362, U01DA055342, U01DA055360, U01DA055350, U01DA055338, U01DA055355, U01DA055363, U01DA055349, U01DA055361, U01DA055316, U01DA055344, U01DA055322, U01DA055369, U01DA055358, U01DA055371, U01DA055359, U01DA055354, U01DA055370,

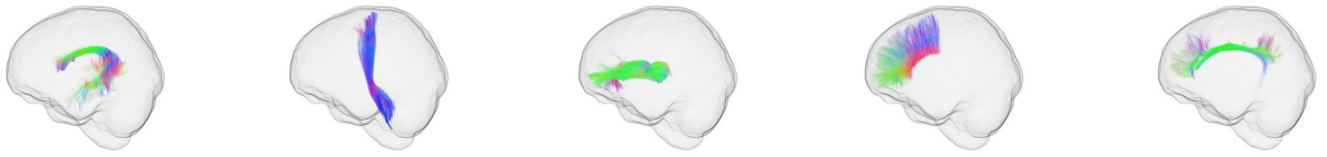
U01DA055347, U01DA055357, U01DA055367, U24DA055325, and U24DA055330. A full list of supporters is available at <https://hbcdstudy.org/federal-partners/>. A full list of participating sites is available at: <https://hbcdstudy.org/recruitment-sites/>. HBCD Study Consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or the HBCD Study Consortium investigators. Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive DevelopmentSM (ABCD) Study (<https://abccdstudy.org/>), held in the NIMH Data Archive (NDA). The ABCD Study is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123 and U24DA041147. A full list of supporters is available at <https://abccdstudy.org/federal-partners.html>. A listing of participating sites and a complete listing of the study investigators can be found at https://abccdstudy.org/consortium_members/. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or ABCD consortium investigators. We thank G. Taylor for assistance in creating some of the pictorial representations that were used in parts of the figures. HABS-HD comes from UNT Health Fort Worth, with the following primary investigators: Sid E. O'Bryant, Kristine Yaffe, Arthur Toga, Robert Rissman and Leigh Johnson; and the HABS-HD investigators: Meredith Braskie, Kevin King, James R. Hall, Melissa Petersen, Raymond Palmer, Robert Barber, Yonggang Shi, Fan Zhang, Rajesh Nandy, Roderick McCol, David Mason, Bradley Christian, Nicole Phillips, Stephanie Large, Joe Lee, Badri Vardarajan, Monica Rivera Mindt, Amrita Cheema, Lisa Barnes, Mark Mapstone, Annie Cohen, Amy Kind, Ozioma Okonkwo, Raul Vintimilla, Zhengyang Zhou, Michael Donohue, Rema Raman, Matthew Borzage, Michelle Mielke, Beau Ances, Ganesh Babulal, Jorge Llibre-Guerra, Carl Hill and Rocky Vig. Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf. Further information on datasets can be found in the Supplementary Acknowledgements.

Author contributions M.E.K. performed the bulk of the data processing and organization, performed the QC, conducted analyses and data interpretation, created figures, created the software for OOS alignment and the associated Docker image with the brain charts, created the Docker image for the postprocessing pipeline, and wrote and edited the manuscript and supplementary information. P.K. created the container for the preprocessing pipeline. K.G.S. provided a supervisory role in the direction of the manuscript and helped design the study. B.A.L. provided a supervisory role in the direction of the manuscript and substantially supported the computational infrastructure and data storage resources used in this study. L.T.D., J.C.G. and K.G.S. provided substantial feedback and editing of the manuscript. C.G., K.R., P.K., N.R.N., G.R. and Z.L. helped with the processing and preprocessing of the data and figure design. S.N.V. and P.Z. provided direction on the design of the statistical analyses. C.G., K.R., N.R.N., P.K., S.B., G.R., D.A., T.J.H., A.L.J., V.L.M., A.R., D.J.E., S.M.R., L.L.B.-H., L.C., L.A.B., M.A.D., T.Q.N., K.L.H., Y.N., S.V.-B., C.J.C., L.T.D., Z.L., S.N.V., P.Z., J.C.G. and B.A.L. reviewed and provided feedback and revisions on the manuscript. M.E.K., D.A., T.J.H., A.L.J., V.L.M., A.R., D.J.E., S.M.R., L.L.B.-H., L.C., L.A.B., M.A.D., T.Q.N., K.L.H., Y.N., S.V.-B., C.J.C., B.A.L. and K.G.S. made substantial contributions to the data acquisition.

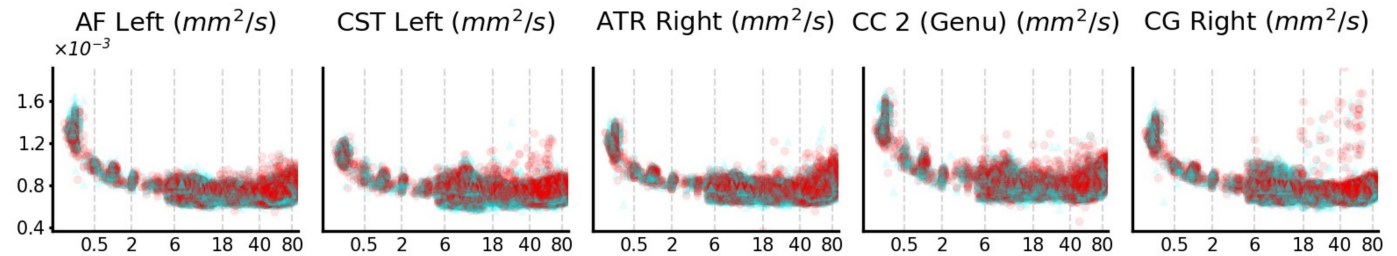
Competing interests T.J.H. serves on the scientific advisory board for Vivid Genomics, serves as a consultant for Circular Genomics, serves as deputy editor of Alzheimer's and Dementia: Translational Research and Clinical Intervention and serves as Senior Associate Editor for Alzheimer's and Dementia. S.M.R. is an employee of the Intramural Research Program of the NIA/NIH; the BLSA studies and investigators are funded by the NIA. K.L.H. participated in the Federation of Associations in Behavioral and Brain Sciences Hill Day in May 2025, meeting with congressional representatives to advocate for continued National Science Foundation (NSF) funding as part of the Coalition for National Science Funding efforts celebrating 75 years of NSF support. A.L.J. serves as a member of the advisory board for Lantheus—Diagnostic and Therapeutic Innovations, Billerica, MA, serves as chair of the Observational Study Monitoring Board, Diverse-VCID: White Matter Lesion Etiology of Dementia in Diverse Populations (Diverse-VCID) Study, Bethesda, MD, is a member of the Observational Study Monitoring Board, Determinants of Incident Stroke Cognitive Outcomes and Vascular Effects on Recovery (DISCOVERY) Study, Bethesda, MD, is a member of the External Advisory Board, Kansas Alzheimer's Disease Core Center (P30), Kansas City, KS, is a member of the Scientific Advisory Committee for Paul B. Beeson Emerging Leaders Career Development Program, John A. Hartford Foundation and American Federations for Aging Research, New York, NY, is a member of the External Advisory Committee for the Clin-STAR Coordinating Center, American Federation for Aging Research, New York, NY, and is a member of the Alzheimer's Disease and Related Dementia Advisory Council, State of Tennessee. The other authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10454-2>.
Correspondence and requests for materials should be addressed to Michael E. Kim, Bennett A. Landman or Kurt G. Schilling.
Peer review information Nature thanks Catherine Lebel, Michel Thiebaut de Schotten and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer review reports are available.
Reprints and permissions information is available at <http://www.nature.com/reprints>.

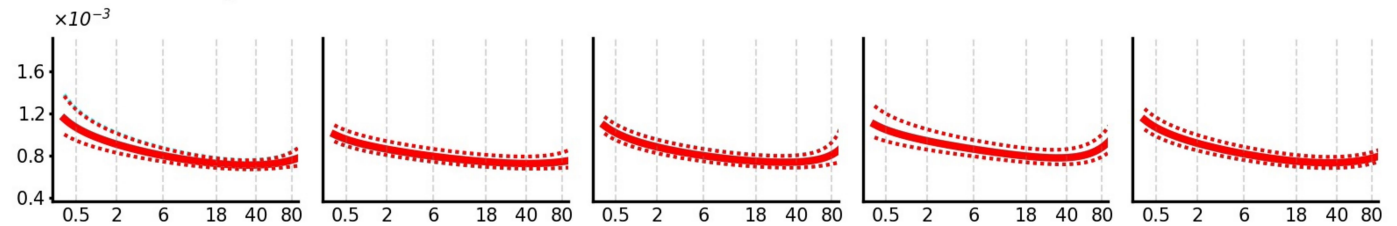
A White Matter Tracts



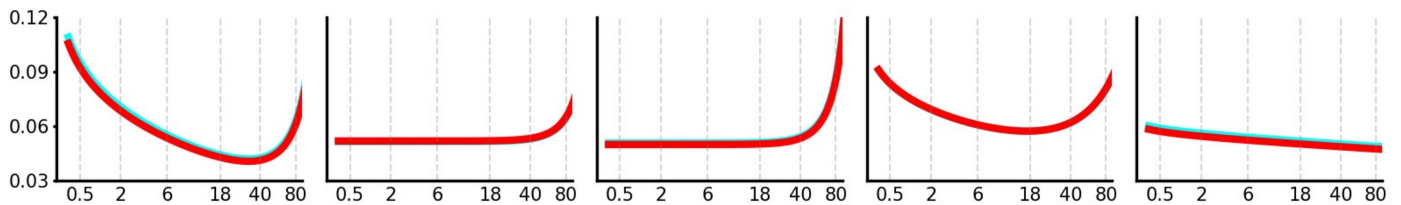
B Tract Mean MD



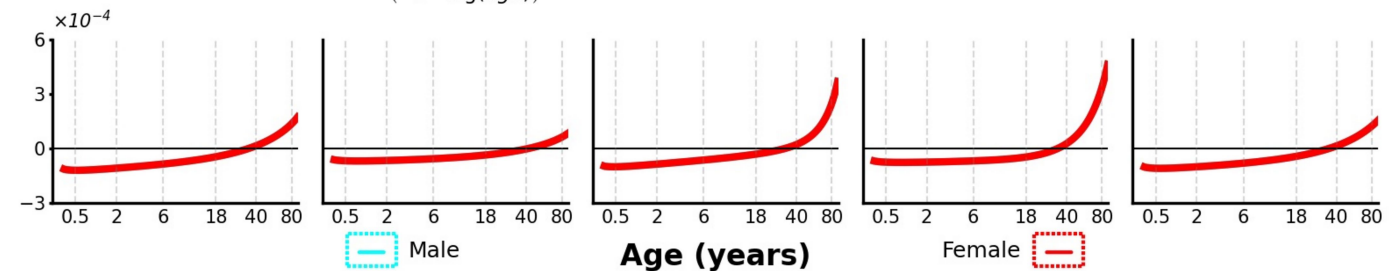
C Normative Trajectories



D Normalized Quantile Ranges



E Normative Rate of Growth ($d/d_{\log(\text{age})}$)

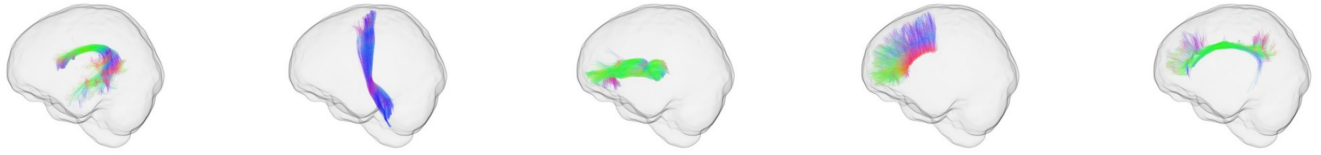


Extended Data Fig. 1 | MD microstructural brain charts. Microstructural brain charts (MD) for five different WM tracts show tract-wise variability and varying developmental milestones across the human lifespan. A.) From left to right, left arcuate fasciculus; left corticospinal tract, right anterior thalamic radiation; genu of the corpus callosum; and the right cingulum. B.) Raw datapoints span the entirety of the lifespan at high density. C.) Most normative trajectories of MD for different pathways decrease in value during development, reach a minimum in adulthood, and then increase at different rates in aging. For all pathways represented, the male trajectories (blue) sit at nearly equal

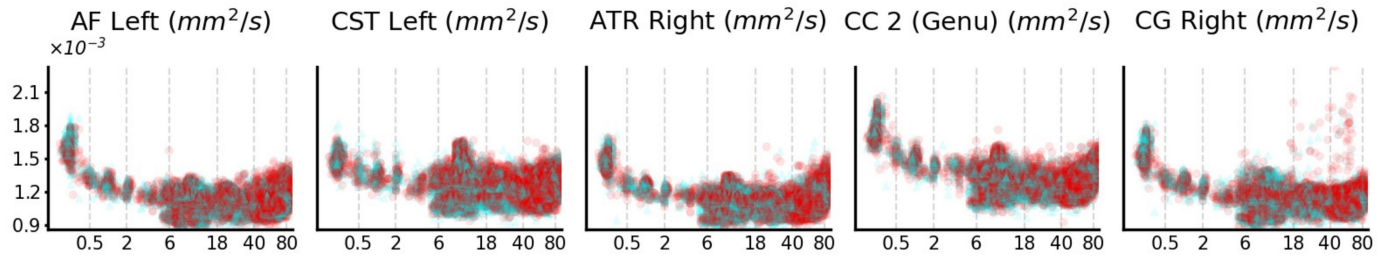
values to the female trajectories (red). D.) Normalized quantile differences indicate that microstructural variability in MD increases at the end of the human lifespan and is lowest in middle age. Variability in MD during development appears to be tract dependent, with some tracts showing lowest variability in infancy and others decreasing to a minimum in adulthood. E.) The normative rate of change ($d/d_{\log(\text{age})}$) of MD for the represented tracts. MD of tracts reach minima at different ages in the lifespan, indicating tract-specific developmental patterns.

Article

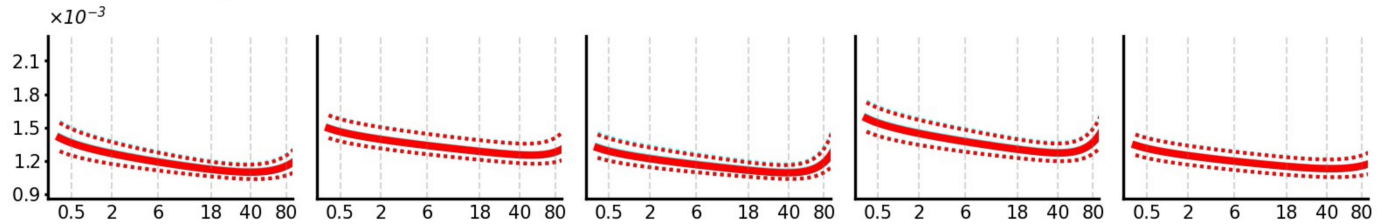
A White Matter Tracts



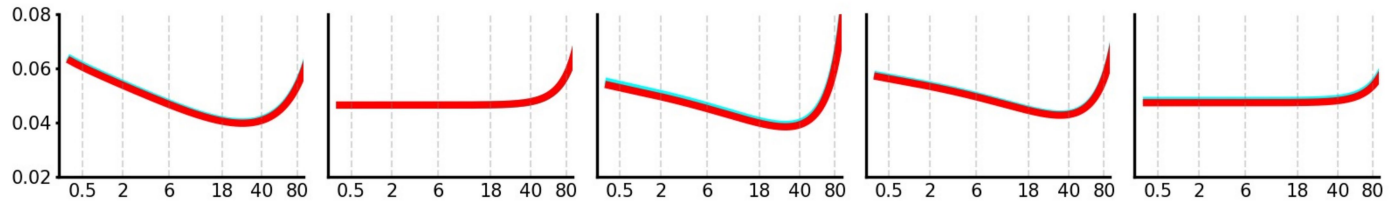
B Tract Mean AD



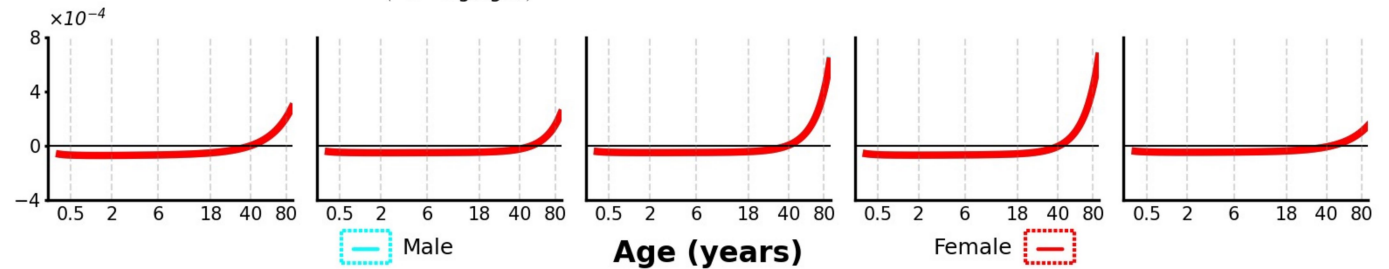
C Normative Trajectories



D Normalized Quantile Ranges



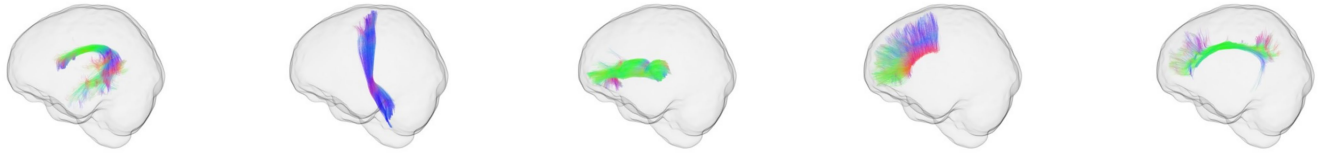
E Normative Rate of Growth ($d/d_{\log(\text{age})}$)



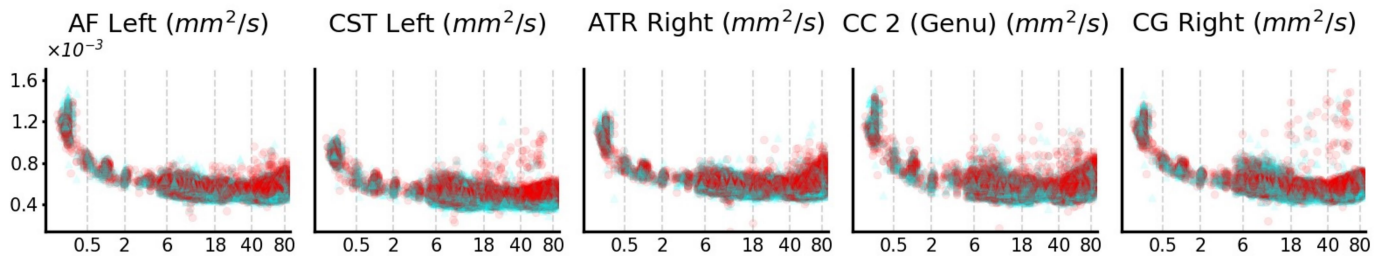
Extended Data Fig. 2 | AD microstructural brain charts. Microstructural brain charts (AD) for five different WM tracts show tract-wise variability and varying developmental milestones across the human lifespan. A.) From left to right, left arcuate fasciculus; left corticospinal tract, right anterior thalamic radiation; genu of the corpus callosum; and the right cingulum. B.) Raw datapoints span the entirety of the lifespan at high density. C.) Normative trajectories of AD for different pathways decrease in value during development, reach a minimum in adulthood, and then increase at different rates in aging.

For all pathways represented, the male trajectories (blue) sit slightly above the female trajectories (red). D.) Normalized quantile differences indicate that microstructural variability in AD increases at the end of the human lifespan and is lowest in middle age. Variability in AD during development appears to be tract dependent, with some tracts showing lowest variability in infancy and others decreasing to a minimum in adulthood. E.) The normative rate of change ($d/d_{\log(\text{age})}$) of AD for the represented tracts. AD of tracts reach minima at different ages in the lifespan, indicating tract-specific developmental patterns.

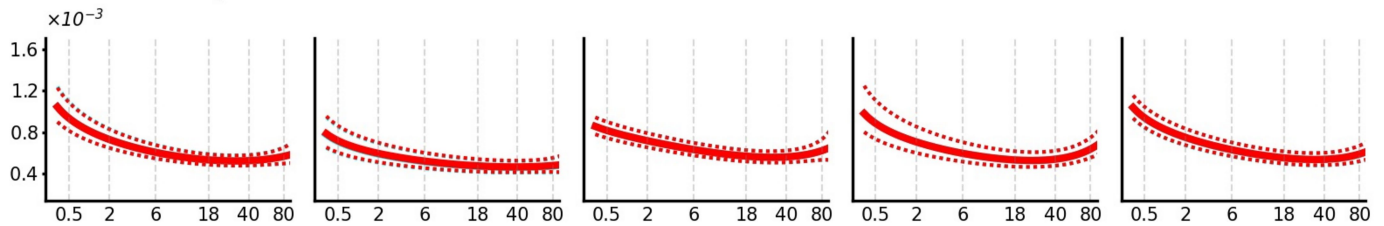
A White Matter Tracts



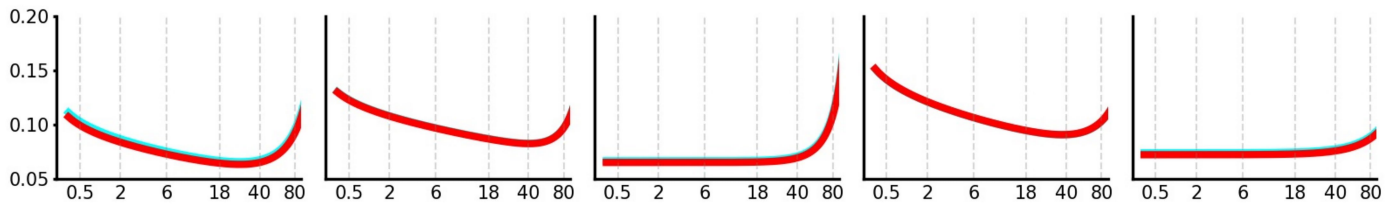
B Tract Mean RD



C Normative Trajectories



D Normalized Quantile Ranges



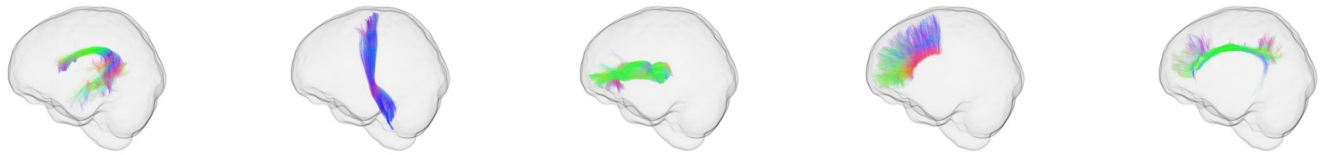
E Normative Rate of Growth ($d/d_{\log(\text{age})}$)



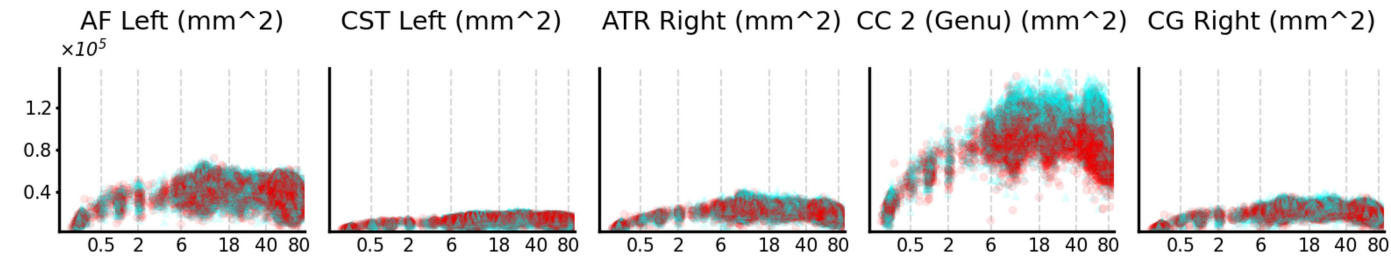
Extended Data Fig. 3 | RD microstructural brain charts. Microstructural brain charts (RD) for five different WM tracts show tract-wise variability and varying developmental milestones across the human lifespan. A.) From left to right, left arcuate fasciculus; left corticospinal tract, right anterior thalamic radiation; genu of the corpus callosum; and the right cingulum. B.) Raw datapoints span the entirety of the lifespan at high density. C.) Normative trajectories of RD for different pathways typically decrease in value during development, reach a minimum in adulthood, and then increase at different

rates in aging. For all pathways represented, the female trajectories (red) sit slightly above the male trajectories (blue). D.) Normalized quantile differences indicate that microstructural variability in RD increases at the end of the human lifespan and is lowest in middle age. RD variability appears to be tract dependent, with some tracts showing lowest variability in infancy and others increasing steadily across the lifespan. E.) The normative rate of change ($d/d_{\log(\text{age})}$) of RD for the represented tracts. RD of tracts reach minima at different ages in the lifespan, indicating tract-specific developmental patterns.

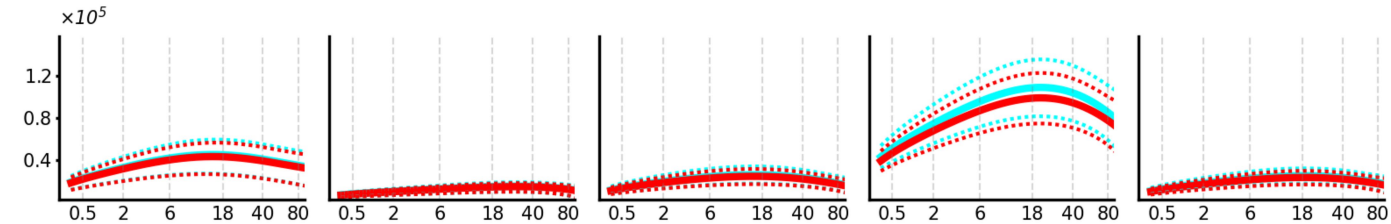
A White Matter Tracts



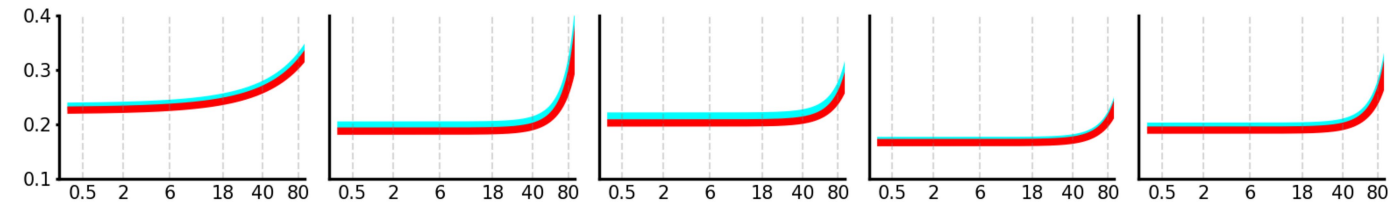
B Tract Surface Area



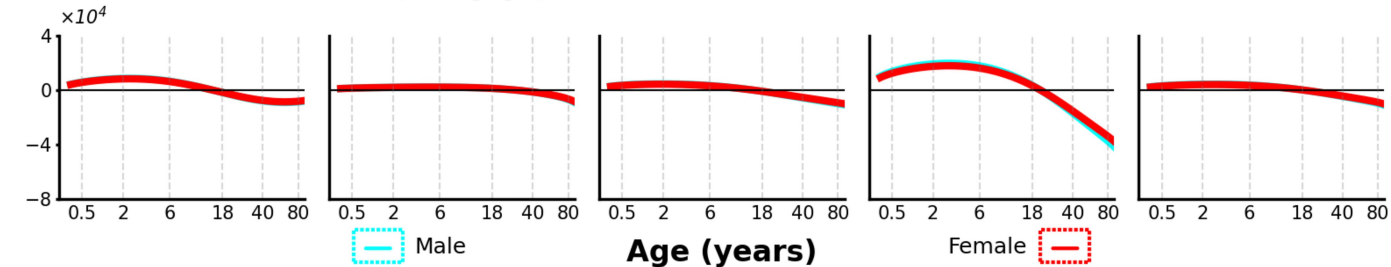
C Normative Trajectories



D Normalized Quantile Ranges



E Normative Rate of Growth ($d/d_{\log(\text{age})}$)

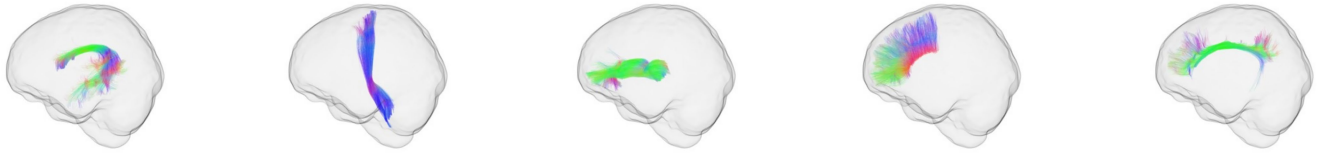


Extended Data Fig. 4 | Surface area macrostructural brain charts.

Macrostructural brain charts (surface area) for five different WM tracts show tract-wise variability and varying developmental milestones across the human lifespan. A.) From left to right, left arcuate fasciculus; left corticospinal tract, right anterior thalamic radiation; genu of the corpus callosum; and the right cingulum. B.) Raw datapoints span the entirety of the lifespan at high density. C.) Normative trajectories of surface area for different pathways increase in value during development, reach a maximum in adulthood, and then decrease

at different rates in aging. For all pathways represented, the male trajectories (blue) sit slightly above the female trajectories (red). D.) Normalized quantile differences indicate that macrostructural variability in tract surface area increases at the end of the human lifespan and is lowest in middle age. Variability in the male trajectories is consistently greater than that of the female trajectories. E.) The normative rate of change ($d/d_{\log(\text{age})}$) of surface area for the represented tracts. Surface area of tracts reach maxima at different ages in the lifespan, indicating tract-specific developmental patterns.

A White Matter Tracts



B Tract Average Length

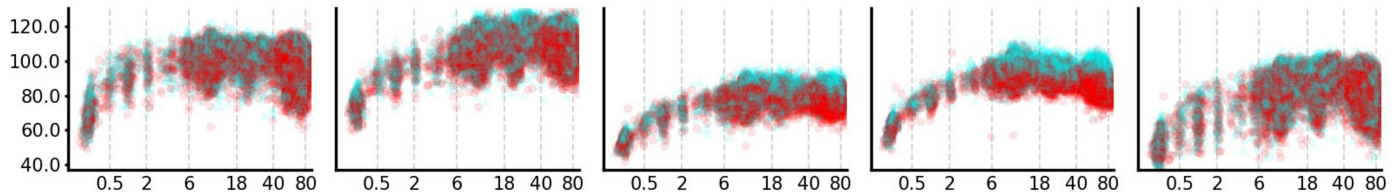
AF Left (mm)

CST Left (mm)

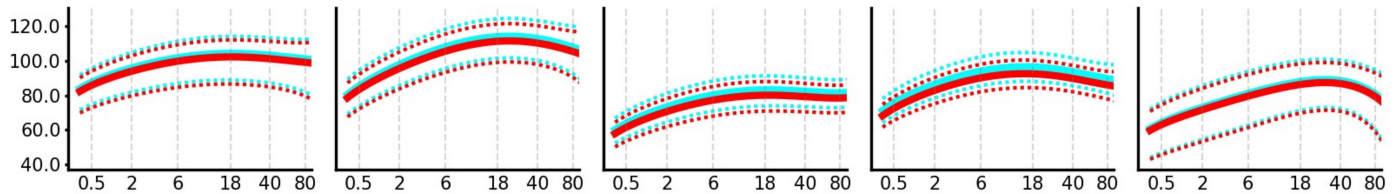
ATR Right (mm)

CC 2 (Genu) (mm)

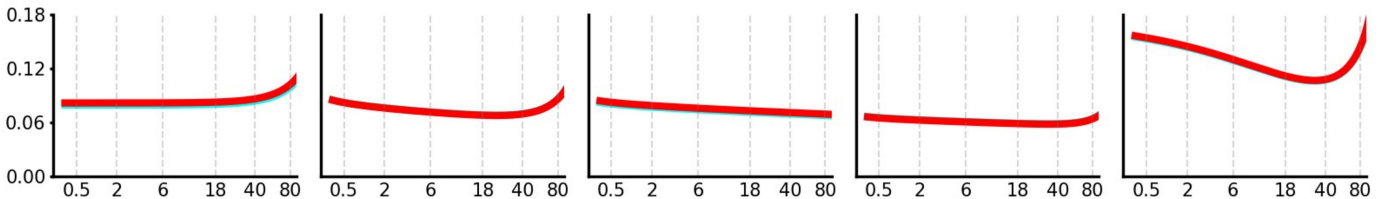
CG Right (mm)



C Normative Trajectories



D Normalized Quantile Ranges



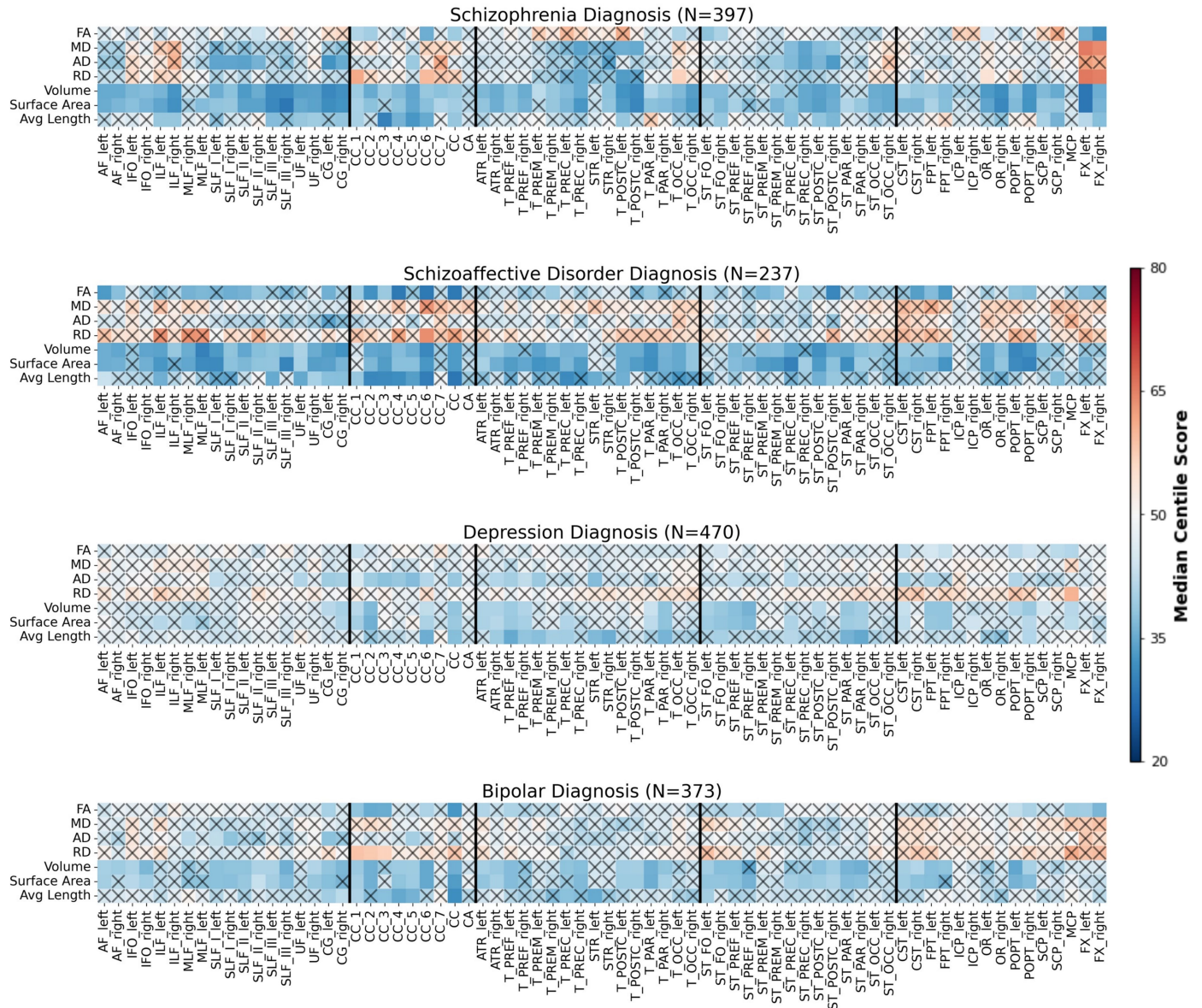
E Normative Rate of Growth ($d/d_{\log(\text{age})}$)



Extended Data Fig. 5 | Average length macrostructural brain charts.

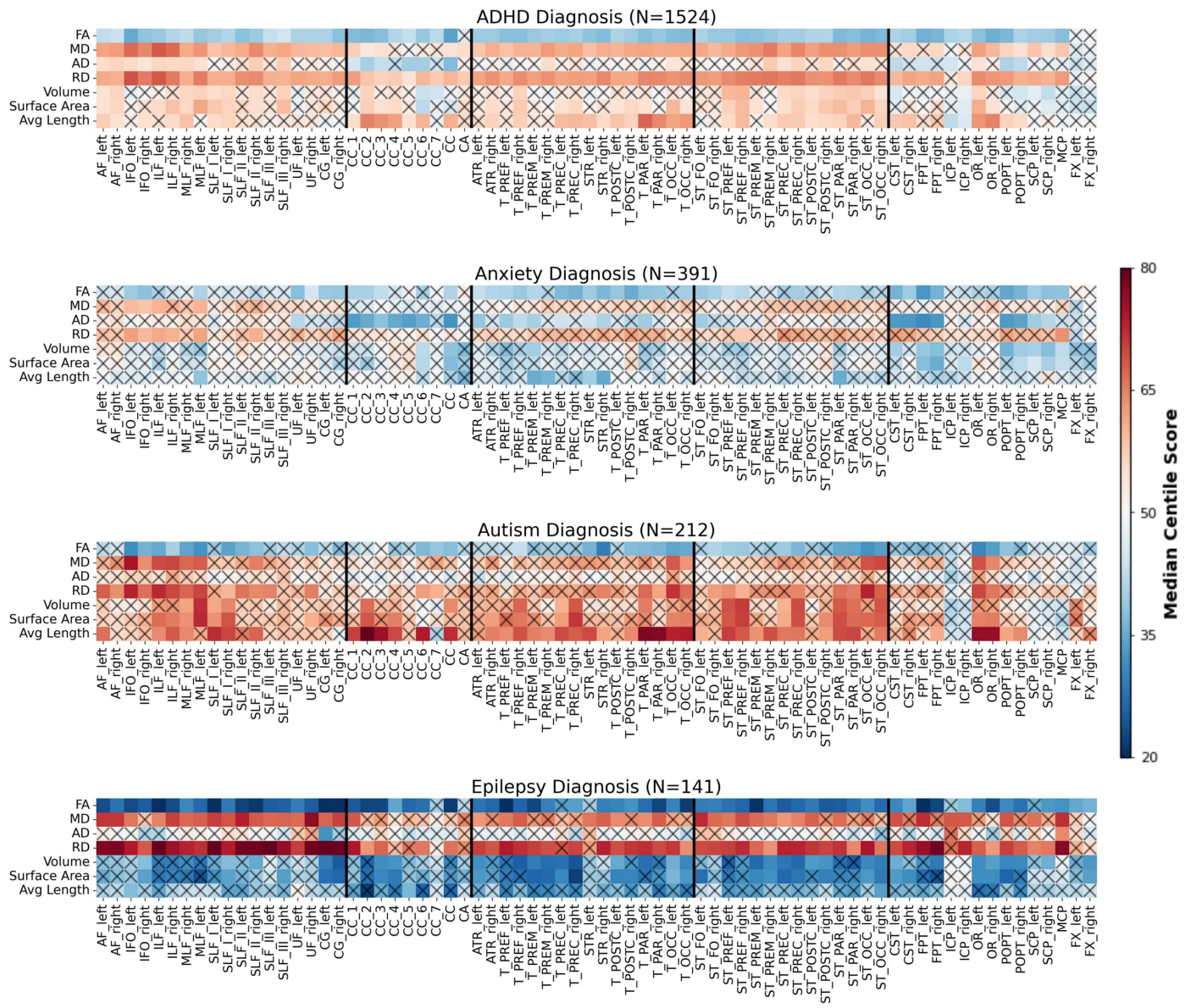
Macrostructural brain charts (average length) for five different WM tracts show tract-wise variability and varying developmental milestones across the human lifespan. A.) From left to right, left arcuate fasciculus; left corticospinal tract, right anterior thalamic radiation; genu of the corpus callosum; and the right cingulum. B.) Raw datapoints span the entirety of the lifespan at high density. C.) Normative trajectories of average length for different represented pathways increase in value during development, reach a maximum in adulthood, and then decrease at different rates in aging. For all pathways represented, the male

trajectories (blue) sit slightly above the female trajectories (red). D.) Normalized quantile differences indicate that macrostructural variability is very tract-dependent. Some tracts have the highest variability at the beginning of the lifespan and lowest at the end, whereas others have the opposite trend. The change in variability of average length across the lifespan also appears to be tract dependent, with some tracts decreasing somewhat consistently and others showing sharp increases. E.) The normative rate of change ($d/d_{\log(\text{age})}$) of average length for the represented tracts. Average length of tracts reach maxima at different ages in the lifespan, indicating tract-specific developmental patterns.



Extended Data Fig. 6 | Centile score differences across major depressive and psychotic disorders. All major psychoses and major depressive disorder (top to bottom: schizophrenia, schizoaffective disorder, major depressive disorder, and bipolar disorder) showed localized reductions in tract

macrostructure and microstructure under the one-sample two-sided Wilcoxon signed-rank test after Bonferroni correction for multiple comparisons. Gray “X” indicates features where the median centile was not significantly different from the 50th normative centile (significant at $p < 9.92 \times 10^{-5}$).



Extended Data Fig. 7 | Centile score differences across other neurocognitive disorders with significant deviations. For many other clinical cohorts (top to bottom: attention-deficit hyperactivity disorder (ADHD), anxiety, autism spectrum disorder, and epilepsy), we observe either localized deviations in tract microstructure and macrostructure or consistent deviations across all tracts within specific macrostructural and microstructural features. Individuals with epilepsy and ADHD showed substantial deviations in FA and RD across the WM, suggesting diffuse microstructural abnormalities, alongside more localized reductions in volume, length, and surface area in select tracts.

For individuals diagnosed with autism spectrum disorder, we observe increases in macrostructure across tracts, with a subset of tracts showing significant deviations. Finally, we observe sparsely significant deviations in microstructure and macrostructure for individuals diagnosed with anxiety. Gray "X" indicates features where the median centile was not significantly different from the 50th normative centile under a two-sided one-sample Wilcoxon signed-rank test following Bonferroni correction for multiple comparisons (significant at $p < 9.92 \times 10^{-5}$).

Article

Extended Data Table 1 | Participants from each dataset used to create the WM brain charts

Dataset	Scans	Participants	Participants After Quality Control	Neurotypical
ABCD	20951	10607	10311	9000
ADNI	3451	1292	1229	584
AOMIC PIOP1	211	211	204	204
AOMIC PIOP2	226	226	224	224
AOMIC ID1000	925	925	858	858
Ageility	131	131	131	131
BANDA	207	207	207	63
BIOCARD	974	210	186	153
BLSA	5423	1035	995	942
BSNIP1	1400	594	554	197
BSNIP2	1432	730	718	233
Calgary	249	88	88	88
CALM	314	314	313	189
CAMCAN	641	641	613	613
VUMC-ASD	173	164	132	63
CUTTING	1661	638	632	597
dHCP	753	668	285	285
DLBS	955	464	444	444
EBDT	1389	468	455	455
HABSHD	3767	2762	2558	1885
HBCD	546	500	285	285
HBN	2760	2754	2519	1049
HCPA	719	719	718	718
HCPBaby	411	211	197	197
HCPD	621	621	621	621
HCP	1109	1065	1064	1064
ABC-Babies	180	142	85	85
IBIS	494	272	217	167
ICBM	264	192	185	185
Lexical	113	113	112	112
MAP	1579	589	393	273
MARS	347	184	172	143
MASiVar	118	83	83	83
MORGAN	414	256	254	110
NACC	1768	1353	677	485
NKI	2224	1289	1205	840
PING	762	762	695	695
QTAB	717	415	414	414
ROS	127	77	46	35
SCAN	303	288	260	188
SWU	460	234	231	231
TempleSocial	108	108	103	103
UCLA	262	262	248	116
UKBB	10425	10425	8509	8509
UPennRisk	251	152	151	151
UTAustin579	376	239	237	237
VMAP_2.0	342	327	266	266
VMAP	1239	328	310	160
TN Alz Project	194	168	159	58
WRAP	570	343	338	332
Total	75036	46846	41891	35120

Extended Data Table 2 | The 72 TractSeg⁵¹ tract definitions with the corresponding tract groups

Tract Name	Tract Description	Tract Group
AF_left	Arcuate Fasciculus Left	Association
AF_right	Arcuate Fasciculus Right	Association
IFO_left	Inferior Occipito-Frontal Fascicle Left	Association
IFO_right	Inferior Occipito-Frontal Fascicle Right	Association
ILF_left	Inferior Longitudinal Fascicle Left	Association
ILF_right	Inferior Longitudinal Fascicle Right	Association
MLF_right	Middle Longitudinal Fascicle Right	Association
MLF_left	Middle Longitudinal Fascicle Left	Association
SLF_I_left	Superior Longitudinal Fascicle I Left	Association
SLF_I_right	Superior Longitudinal Fascicle I Right	Association
SLF_II_left	Superior Longitudinal Fascicle II Left	Association
SLF_II_right	Superior Longitudinal Fascicle II Right	Association
SLF_III_left	Superior Longitudinal Fascicle III Left	Association
SLF_III_right	Superior Longitudinal Fascicle III Right	Association
UF_left	Uncinate Fasciculus Left	Association
UF_right	Uncinate Fasciculus Right	Association
CG_left	Cingulum Left	Association
CG_right	Cingulum Right	Association
CC_1	Corpus Callosum 1	Commissural
CC_2	Corpus Callosum 2	Commissural
CC_3	Corpus Callosum 3	Commissural
CC_4	Corpus Callosum 4	Commissural
CC_5	Corpus Callosum 5	Commissural
CC_6	Corpus Callosum 6	Commissural
CC_7	Corpus Callosum 7	Commissural
CC	Corpus Callosum	Commissural
CA	Anterior Commissure	Commissural
ATR_left	Anterior Thalamic Radiation Left	Thalamic
ATR_right	Anterior Thalamic Radiation Right	Thalamic
T_PREF_left	Thalamic-Prefrontal Left	Thalamic
T_PREF_right	Thalamic-Prefrontal Right	Thalamic
T_PREM_left	Thalamic-Premotor Left	Thalamic
T_PREM_right	Thalamic-Premotor Right	Thalamic
T_PREC_left	Thalamic-Precentral Left	Thalamic
T_PREC_right	Thalamic-Precentral Right	Thalamic
STR_left	Superior Thalamic Radiation Left	Thalamic
STR_right	Superior Thalamic Radiation Right	Thalamic
T_POSTC_left	Thalamic-Postcentral Left	Thalamic
T_POSTC_right	Thalamic-Postcentral Right	Thalamic
T_PAR_left	Thalamic-Parietal Left	Thalamic
T_PAR_right	Thalamic-Parietal Right	Thalamic
T_OCC_left	Thalamic-Occipital Left	Thalamic
T_OCC_right	Thalamic-Occipital Right	Thalamic
ST_FO_left	Striato-Fronto-Orbital Left	Striatal
ST_FO_right	Striato-Fronto-Orbital Right	Striatal
ST_PREF_left	Striato-Prefrontal Left	Striatal
ST_PREF_right	Striato-Prefrontal Right	Striatal
ST_PREM_left	Striato-Premotor Left	Striatal
ST_PREM_right	Striato-Premotor Right	Striatal
ST_PREC_left	Striato-Precentral Left	Striatal
ST_PREC_right	Striato-Precentral Right	Striatal
ST_POSTC_left	Striato-Postcentral Left	Striatal
ST_POSTC_right	Striato-Postcentral Right	Striatal
ST_PAR_left	Striato-Parietal Left	Striatal
ST_PAR_right	Striato-Parietal Right	Striatal
ST_OCC_left	Striato-Occipital Left	Striatal
ST_OCC_right	Striato-Occipital Right	Striatal
CST_left	Corticospinal Tract Left	Projection
CST_right	Corticospinal Tract Right	Projection
FPT_left	Frontopontine Tract Left	Projection
FPT_right	Frontopontine Tract Right	Projection
ICP_left	Inferior Cerebellar Peduncle Left	Projection
ICP_right	Inferior Cerebellar Peduncle Right	Projection
OR_left	Optic Radiation Left	Projection
OR_right	Optic Radiation Right	Projection
POPT_left	Parieto-Occipito-Temporal Left	Projection
POPT_right	Parieto-Occipito-Temporal Right	Projection
SCP_left	Superior Cerebellar Peduncle Left	Projection
SCP_right	Superior Cerebellar Peduncle Right	Projection
MCP	Middle Cerebellar Peduncle	Projection
FX_left	Fornix Left	Projection
FX_right	Fornix Right	Projection

Note that for fitting out-of-sample datasets, the values in the “Tract Name” column should be used for the CSV when performing out-of-sample alignment.

Extended Data Table 3 | Participants classified as cognitively abnormal for each dataset and diagnosis

Dataset	Autism	ADHD	Alzheimer's	Anxiety	Bipolar	CN*	Defiance/ Conduct Disorder	Depression	Dyslexia	Epilepsy	MCI	Schizoaffective Disorder	Schizophrenia	Social Anxiety	Substance Abuse	Suicidality and Self- Injury	Other	Total
ABCD		619		38	162		89	148					33	40	25	60	97	1311
ADNI			206 (5)								439 (7)							645 (12)
BANDA				79				65										144
BIOCARD			3			1					29							33
BLSA			17 (4)			4					32 (1)							53 (5)
BSNIP1				1	61 (1)			53 (2)				71 (3)	125 (3)		26 (2)		20 (1)	357 (12)
BSNIP2					101 (4)			10				166 (4)	192 (1)		15		7	491 (9)
CALM	20	66							22								16	124
VUMC-ASD	69																	69
CUTTING																	35	35
HABSHD			36 (1)			34					603 (16)							673 (17)
HBN	73	752 (4)		246	3		37 (1)	144 (1)	69				1	52			93 (2)	1470 (8)
IBIS	50 (2)																	50 (2)
MAP			14 (1)			1					105 (2)							120 (3)
MARS											29 (1)							29 (1)
MORGAN										141							3	144
NACC			160 (3)								32 (3)							192 (6)
NKI		47		27				50							143		98	365
ROS											11							11
SCAN			21 (1)								51 (2)							72 (3)
UCLA		40			46								46 (1)					132 (1)
VMAP			33			1					116 (2)							150 (2)
TN Alz Proj			43 (2)								58 (3)							101 (5)
WRAP			2								4							6
Total	212 (2)	1524 (4)	535 (17)	391	373 (5)	41	126 (1)	470 (3)	91	141	1509 (37)	237 (7)	397 (5)	92	209 (2)	60	369 (3)	6777 (87)

Values in parentheses denote the number dropped from the +/-4 standard deviation outlier removal detailed in the Methods. CN* denotes individuals diagnosed as cognitively unimpaired at scan time, but were later diagnosed as having some degree of cognitive impairment (including dementia).

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give <i>P</i> values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No custom software was used by the investigators to collect data for this study.
Data analysis	Code to both build and use the white matter brain charts described in this study is available as a Docker image at the following Zenodo repository: https://zenodo.org/records/15367425. We have included documentation, including extensive tutorials and instructions, on how researchers can both use our brain charts and create their own using our code. Furthermore, we have publicly released the post-processing pipeline as a single containerized image with instructions on how to run the code (with the preprocessing container already available) in order to facilitate researchers being able to use our released brain charts- intructions: https://zenodo.org/records/17144460; post-processing container: https://hub.docker.com/r/kimm58/wm_lifespan_processing; pre-processing container: https://doi.org/10.5281/zenodo.14058394. All analysis visualizations and statistics were generated using Python version 3.9.20. Notable python libraries used are pandas (v2.2.3), numpy (v2.0.2), seaborn (v0.13.2), matplotlib (v3.9.2), rpy2 (v3.5.11), and scipy (v1.13.1). Heatmaps to visualize large matrices were created using "heatmap" in seaborn. Density plots were created using "violin_plot" in seaborn. Datapoints and trajectories across the lifespan were plotted using "scatterplot" from seaborn and "plot" in matplotlib respectively, with the "age" axis scaled logarithmically. Histograms were created using the "hist" function in matplotlib. Linear regressions with confidence intervals were created using "regplot" in seaborn. Graphical representations of white matter tracts in the brain were created using mrview from MRtrix3. Two-sided, one-sample Wilcoxon tests were run using "scipy.stats.wilcoxon" from scipy. Mann Whitney.16-U tests were run using "scipy.stats.mannwhitneyu". For creating 3D qualitative visualizations of white matter tracts, we use the mrview tool inside the MRtrix3 (version 3.0.4). Alternatively, for qualitative region of interest 3D visualizations, the OpenDIVE visualization library (v0.3.0, https://github.com/MASILab/open_dive) was used, where a different Python version, 3.10.18, was used in these visualization instances for library compatibility purposes. For Figure 6C, the graphical head is a combination of a FreeSurfer (v7.2) cortical segmentation of a brain scan (which was made by the authors) and graphics (gears, head outline) that were hand-sketched using graphic design software (krita v5.2.16). For processing of the MRI data, we use several open source tools and

pipelines, such as FSL (v6.0.4), MRtrix3 (v3.0.4), ANTs (v2.5.0), scilpy (v1.5.0), synbo-DISCO (v3.1), FreeSurfer (v7.2), and TractSeg (v2.8). A more detailed description of the data processing used can be found in the methods section of the manuscript.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about availability of data

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All derived data and associated demographic information from publicly available datasets, as well as from datasets for which the Data Use Agreement (DUA) permits sharing, have been made accessible at <https://zenodo.org/records/18891847>. For datasets where the DUA does not permit redistribution of derivative data, these materials have been returned to or shared directly with the respective data custodians.

Data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) are available upon request from <https://adni.loni.usc.edu/>. Data from the AOMIC-PIOP1 dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds002785/versions/2.0.0>. Data from the AOMIC-PIOP2 dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds002790/versions/2.0.0>. Data from the AOMIC-ID1000 dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds003097/versions/1.2.1>. Data from the Boston Adolescent Neuroimaging of Depression and Anxiety (BANDA) dataset are available upon request from <https://www.humanconnectome.org/study/connectomes-related-anxiety-depression>. Data from BIOCARD are available upon request after filling out a data use application: <https://www.gaaindata.org/partner/BIOCARD>. Data from the Baltimore Longitudinal Study of Aging (BLSA) are available upon request from <https://www.blsa.nih.gov/>. Data from the Calgary Preschool MRI Dataset are freely available for download at: <https://doi.org/10.17605/OSF.IO/AXZ5R>. Data from the Centre for Attention Learning and Memory (CALM) dataset are available upon request from <https://calm.mrc-cbu.cam.ac.uk/researchers/>. Data from the Cambridge Center for Ageing Neuroscience (CAMCAN) dataset are available upon request from <https://camcan-archive.mrc-cbu.cam.ac.uk/dataaccess/>. Data from the Dallas Lifespan Brain Study (DLBS) dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds004856/versions/1.2.0>. Data from the Health & Aging Brain Study - Health Disparities (HABS-HD) dataset are available upon request from <https://apps.unthsc.edu/itr/reports>. Imaging data and basic demographic information for the Healthy Brain Network (HBN) dataset are freely available to download from https://fcon_1000.projects.nitrc.org/indi/cmi_healthy_brain_network/. Phenotypic data are available upon request by filling out a data use agreement: https://fcon_1000.projects.nitrc.org/indi/cmi_healthy_brain_network/Phenotypic.html. Data from the Human Connectome Project - Aging (HCPA) dataset are available upon request from <https://www.humanconnectome.org/study/hcp-lifespan-aging>. Data from the Lifespan Baby Connectome Project (HCPBaby) dataset are available upon request <https://www.humanconnectome.org/study/lifespan-baby-connectome-project>. Data from the Human Connectome Project - Development (HCPD) dataset are available upon request from <https://www.humanconnectome.org/study/hcp-lifespan-development>. Data from the Human Connectome Project - Young Adult (HCP) dataset are freely available for download from <https://www.humanconnectome.org/study/hcp-young-adult>. Data from the Infant Brain Imaging Study (IBIS) are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=19. Data from the International Consortium for Brain Mapping (ICBM) dataset are available upon request from www.loni.usc.edu/ICBM. Data from the Longitudinal Brain Correlates of Multisensory Lexical Processing in Children study (shortened to Lexical in this manuscript) are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds001894/versions/1.4.2>. Data from the Memory and Aging Project (MAP), Religious Orders Study (ROS), and the Minority Aging Research Study (MARS) datasets are available upon request from <https://www.radc.rush.edu/>. Data from the Multisite, Multiscanner, and Multisubject Acquisitions for Studying Variability in Diffusion Weighted Magnetic Resonance Imaging (MASiVar) dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds003416/versions/2.0.2>. Data from the National Alzheimer's Coordinating Center (NACC) and the Standardized Centralized Alzheimer's & Related Dementias Neuroimaging (SCAN) are available upon request from <https://nacccdata.org/requesting-data/data-request-process>. Imaging data and basic demographic information for the Nathan Kline Institute - Rockland Sample (NKI) dataset are freely available to download from https://fcon_1000.projects.nitrc.org/indi/enhanced/sharing_neuro.html. Phenotypic data are available upon request by filling out a data use agreement: https://fcon_1000.projects.nitrc.org/indi/enhanced/sharing_phenotypic.html. Data from the Pediatric Imaging, Neurocognition, and Genetics dataset (PING) are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=2607. Data from the Queensland Twin Adolescent Brain (QTAB) dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds004146/versions/1.0.4>. Data from the UCLA Consortium for Neuropsychiatric Phenomics LA5c Study (UCLA) dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds000030/versions/1.0.0>. Data from the Southwest University (SWU) Longitudinal Imaging Multimodal dataset are freely available for download here: https://fcon_1000.projects.nitrc.org/indi/retro/southwestuni_giu_index.html. Data from the Social Reward and Nonsocial Reward Processing Across the Adult Lifespan: An Interim Multi-echo fMRI and Diffusion Dataset (referred to as TempleSocial in this manuscript) are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds005123/versions/1.1.3>. Data from UK Biobank (UKBB) are available upon request from <https://www.ukbiobank.ac.uk/>. Data from the UPennRisk dataset are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds002843/versions/1.0.1>. Data from the dataset titled "A longitudinal neuroimaging dataset on language processing in children ages 5, 7, and 9 years old" (referred to as UTAustin579 in this manuscript) are freely available for download on OpenNeuro: <https://openneuro.org/datasets/ds003604/versions/1.0.7>. Data from the Vanderbilt Memory and Aging Project (VMAP_JEFFERSON, VMAP_2.0, TN Aging Project) are available for download upon request from <https://vmacdata.org/vmap/data-requests>. Data from the Wisconsin Registry for Alzheimer's Prevention (WRAP) are available upon request from <https://wrap.wisc.edu/data-requests-2/>. Data from the HEALTHY Brain and Child Development (HBCD) Study are available for download upon request from (<https://hbcdstudy.org/data-sharing/>). Data from the Adolescent Brain Cognitive Development (ABCD) Study are available for download upon request from (<https://abcdstudy.org/scientists/data-sharing/>). Data used in this manuscript from the Ageility Project (Phase 1 only) are available for download from (<https://www.nitrc.org/projects/age-ility>). Data from the Early Brain Development in Twins (EBDT) dataset are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=2384. Data from the Bipolar & Schizophrenia Consortium for Parsing Intermediate Phenotypes dataset (BSNIP1) and its renewal (BSNIP2) are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=2274 and https://nda.nih.gov/edit_collection.html?id=2165. Data from the Developing Human Connectome Project (dHCP) are available for download from the National Institutes of Mental Health data archive upon request from https://nda.nih.gov/edit_collection.html?id=3955. Vanderbilt University data (MORGAN, BABIES-ABC, CUTTING, VUMC-ASD) subject to third party restrictions. Please contact corresponding authors for data requests.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Sex was used as a covariate of interest in the study and reported according to the source dataset-specific policies. For instances when sex information was unavailable from the individual dataset, reported gender was used as a replacement variable.
Reporting on race, ethnicity, or other socially relevant groupings	No information on race/ethnicity was used for analyses in the manuscript. However, we report the aggregate demographic information regarding race/ethnicity in the Supplemental Information (Supplemental Table SB.T1), where such demographic information was reported by the original individual datasets.
Population characteristics	Age was reported in years, whereas for neonatal datasets it was reported as gestational age. Diagnoses were determined according to the source dataset-specific policies; further information about how individuals were diagnosed for each dataset can be found in the dataset specific references in the Supplemental Information (Supplemental Table SB.T2).
Recruitment	Analyses in the manuscript were based on existing datasets; further information about how individuals were recruited for each dataset can be found in the dataset specific references in the Supplemental Information (Supplemental Table SB.T2).
Ethics oversight	The following statement has been added to the Methods section: The research consists of secondary analyses of de-identified primary datasets. Information regarding informed consent of participants (or guardians) in primary studies can be found in the references in Supplemental Table SB.T2. Secondary analysis of these data was also approved by the Vanderbilt University Medical School Institutional Review Board (IRB #210968).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size was determined based on the number of participant scans for any particular cohort (either typically developing/aging or clinical) that passed the quality control procedures detailed in the methods section. Although no a priori sample size was calculated, we conduct variability assessments of the models (Supplemental Figure SD.F3) in addition to the built-in ML optimization of our models to ensure that data and brain charts are robust.
Data exclusions	As described in the methods, data were excluded from analyses if age, sex, or diagnosis was missing for the participant at the time of scan, or if the data did not pass the quality control procedures.
Replication	All data processing was standardized through the use of version-controlled containerized code for replicability. Furthermore, we have publicly released the post-processing pipeline as a single containerized image with instructions on how to run the code (with the preprocessing container already available) - instructions: https://zenodo.org/records/17144460 ; post-processing container: https://hub.docker.com/repository/docker/kimm58/wm_lifespan_processing/general ; pre-processing container: https://doi.org/10.5281/zenodo.14058394 .
Randomization	Allocation was based on the diagnosis specified by each individual dataset. Participants were assigned to the clinical group that was indicated in the source dataset that they originated from. Typically developing/aging participants were those that were specified as either part of a control group or having no diagnosis present.
Blinding	Blinding was not applicable/not possible for a few reasons. First, we used pre-existing MRI datasets from multiple sources and did not acquire MRI data for the purposes of this study. Second, knowledge of diagnostic status was essential to our study design, as we specifically required identification of typically developing/aging individuals to construct normative models. Furthermore, clinical labels were necessary for evaluating deviations from normative trajectories in clinical groups. As such, blinding to group allocation would have been incompatible with the objectives of the study.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☐	☒ MRI-based neuroimaging

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

Magnetic resonance imaging

Experimental design

Design type	Design specifications were dependent on the previously collected datasets used in this study. However, all scanning sessions used in these analyses required a diffusion-weighted MRI scan.
Design specifications	For data collection, design specifications were dependent on the previously collected datasets used in this study. Some datasets were longitudinal in nature or had repeated scans while others were cross-sectional. Regardless of how many data points were collected, we used only a single scan for each participant.
Behavioral performance measures	Any behavioral performance measures collected for the original datasets used in this study were not used in any analyses.

Acquisition

Imaging type(s)	Diffusion and structural
Field strength	Variable: dependent on the dataset and scanner used for each particular source dataset.
Sequence & imaging parameters	Variable: dependent on the dataset and scanner used for each particular source dataset.
Area of acquisition	Whole brain scan
Diffusion MRI	☒ Used ☐ Not used
Parameters	Variable: dependent on the dataset and scanner used for each particular source dataset.

Preprocessing

Preprocessing software	The preprocessing software used was the PreQual software package developed by Cai et al. (2021), which corrects for susceptibility-, eddy current-, and motion-induced artifacts. PreQual also performs denoising of images and performs slice-wise signal imputation for slices with signal dropout. When available, reverse phase encoding scans were used to correct for susceptibility-induced distortions. Otherwise, a T1-weighted image from the same scanning session was used for the correction with Synbo-DisCo developed by Schilling et al. (2019). Code (as a container) and instructions on running the preprocessing pipeline can be found here: https://doi.org/10.5281/zenodo.14058394
Normalization	For scans where multiple NIFTI files were preprocessed together (e.g. including a reverse phase encoding scan for distortion correction), intensity normalization across NIFTIs was performed in the PreQual package. For creating the normative models,

dataset-specific biases were modeled as covariates to perform harmonization across multiple datasets.

Normalization template

Data were not normalized to a standard space for this study.

Noise and artifact removal

All denoising and image distortion removal was performed on the diffusion-weighted MRI scans using the PreQual preprocessing package. The denoising comes from MRtrix3's dwidenoise (PCA denoising). Susceptibility-, eddy current-, and motion-induced artifacts were corrected using FSL's topup and eddy. Slice-wise signal imputation for slices with signal dropout was also performed by FSL's eddy.

Volume censoring

No volume censoring was used in this study.

Statistical modeling & inference

Model type and settings

Generalized additive models for location, scale, and shape (GAMLSS) were used to create the normative models described in the manuscript, specifying a generalized gamma distribution. Age was modeled non-linearly using fractional polynomials of one to three terms (model selection was done using Bayesian information criterion) whereas sex was modeled linearly. Dataset/batch was modeled using random effect terms for intercept.

Effect(s) tested

Our study was not based on task or stimulus conditions. Instead, we investigated normative variation in white matter for typically developing/aging individuals across the human lifespan using demographic variables of age and sex (with a random intercept for dataset/batch). For clinical groups, we assessed deviations of centile score distributions based on the median distance from the median centile.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☒ Both

Anatomical location(s)

Anatomical locations were determined by Tractseg, a deep learning algorithm that segments white matter bundles in the brain. Cerebral white matter was identified using a white matter mask derived from the FreeSurfer segmentation tool.

Statistic type for inference

We did not include voxel-wise or cluster-wise analyses in this study.

(See [Eklund et al. 2016](#))

Correction

Bonferroni correction, obtained based on the number of tests evaluated for an analysis.

Models & analysis

n/a | Involved in the study

- ☒ ☐ Functional and/or effective connectivity
☒ ☐ Graph analysis
☐ ☒ Multivariate modeling or predictive analysis

Multivariate modeling and predictive analysis

Independent variables were age in years, sex as a binary variable, and dataset/batch as a random effect term. The GAMLSS framework was used to model white matter features (linearly and non-linearly) using these independent variables. Model selection was performed using Bayesian information criterion.